

UNI2U-BENCH: DO UNIFIED MODELS ADVANCE MULTIMODAL UNDERSTANDING?

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ABSTRACT

Unified multimodal models have recently demonstrated strong generative capabilities, yet whether and when generation improves understanding remains unclear. Existing benchmarks lack a systematic exploration of the specific tasks where generation facilitates understanding. To this end, we introduce UniG2U-Bench, a comprehensive benchmark categorizing generation-to-understanding (G2U) evaluation into 7 regimes and 30 subtasks, requiring varying degrees of implicit or explicit visual transformations. Extensive evaluation of over 30 models reveals three core findings: 1) Unified models generally underperform their base Vision-Language Models (VLMs), and Generate-then-Answer (GtA) inference typically degrades performance relative to direct inference. 2) Consistent enhancements emerge in spatial intelligence, visual illusions, or multi-round reasoning subtasks, where enhanced spatial and shape perception, as well as multi-step intermediate image states, prove beneficial. 3) Tasks with similar reasoning structures and models sharing architectures exhibit correlated behaviors, suggesting that generation-understanding coupling induces class-consistent inductive biases over tasks, pretraining data, and model architectures. These findings highlight the necessity for more diverse training data and novel paradigms to fully unlock the potential of unified multimodal modeling.

1 INTRODUCTION

Recent advances in multimodal foundation models have led to a growing interest in unifying understanding and generation within a single architecture [Yang et al. \(2025b\)](#). Unified models have demonstrated impressive performance in image synthesis, editing, and cross-modal generation, suggesting that strong perceptual and semantic understanding serves as a critical foundation for high-quality generation [Zou et al. \(2025\)](#); [Xie et al. \(2025b\)](#); [Li et al. \(2025c\)](#); [Xu et al. \(2026\)](#).

Despite this progress, the interaction between understanding and generation has been studied predominantly in a single direction: understanding improves generation. Whether generation itself can enhance understanding by serving as an external medium for reasoning, verification, or hypothesis construction, remains largely unexplored.

A key reason is the absence of benchmarks that explicitly evaluate understanding *through* generation. Most existing benchmarks for multimodal understanding emphasize recognition, classification, or verbal reasoning over static visual inputs. As a result, strong performance can often be achieved by first converting images into dense textual descriptions and then applying a text-only large language model [Xing et al. \(2025\)](#); [Qiao et al. \(2024\)](#). Consequently, many questions in popular understanding benchmarks can be answered without relying on visual information that is inherently non-linguistic. However, in many complex scenarios—such as drawing auxiliary constructions in geometry, reconstructing spatial layouts, or externalizing intermediate states in puzzles and games—effective understanding cannot be reduced to linguistic abstraction alone. In such cases, generation must serve as an internal or external reasoning mechanism that enables access to visual transformations not captured by textual descriptions.

The rapid evolution of unified multimodal models—such as Bagel, OmniGen, Show-o, and Janus—has proven that generation and understanding can coexist within a shared parameter space. While recent benchmarks like UniMMU [Zou et al. \(2025\)](#) and MME-Unify [Xie et al. \(2025b\)](#) evaluate these models, they primarily assess understanding and generation in isolation (i.e., asking “can it

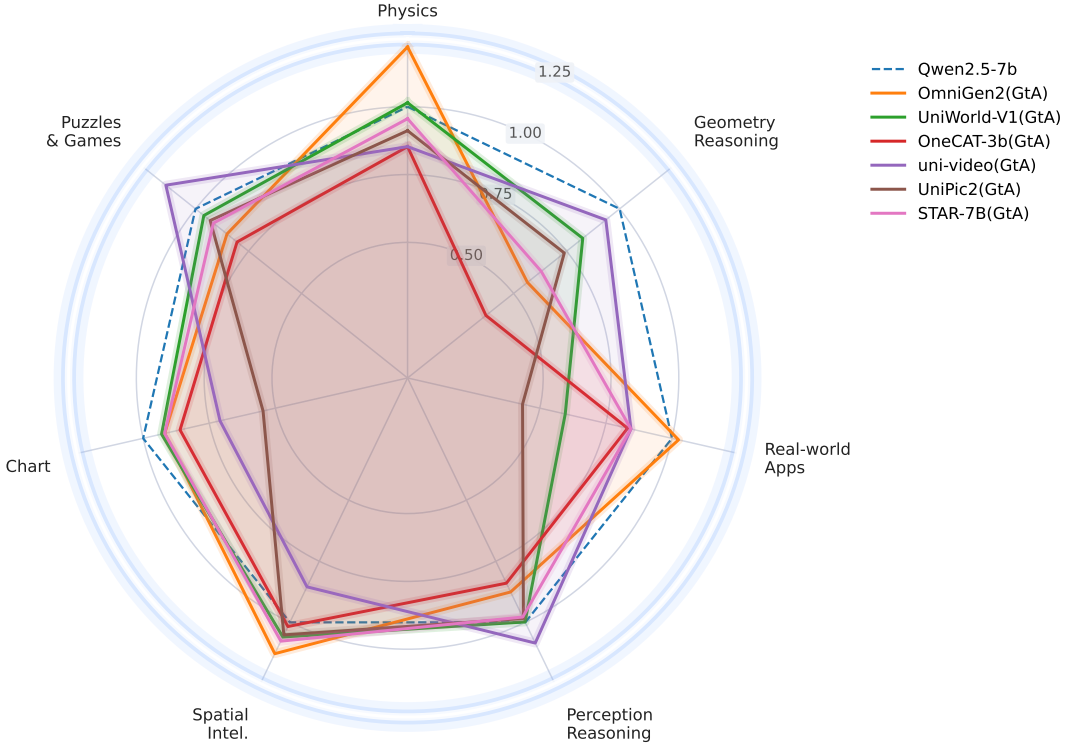


Figure 1: Model Performance Radar Chart

answer?” and “can it draw?” independently). They fail to address whether the generative capacity actively *aids* the understanding process. Furthermore, evaluating these models is challenging due to the confounding variables of model scaling and backbone differences.

To systematically study this, we introduce **UniG2U**, which stands as the most comprehensive understanding benchmark to date for unified models—featuring the largest sample volume, the most diverse task categories, the widest model coverage, and the most complete evaluation framework. Unlike previous benchmarks, UniG2U strictly pairs cutting-edge unified models with their purely discriminative base VLMs under matched inference protocols. By doing so, we cleanly isolate the Generation-to-Understanding (G2U) capability, allowing us to accurately assess whether explicit or implicit visual generation facilitates multimodal reasoning.

We benchmarked over 30 models—encompassing base visual understanding models (VLMs), unified models, and agentic models (which we also define as generalized unified models)—across seven reasoning regimes. Each unified model is evaluated under two distinct inference modes: Direct and Generate-then-Answer (GtA). By rigorously comparing unified models with their base VLMs, we observe a nuanced Generation-to-Understanding (G2U) dynamic, characterized by varying degrees of performance improvements and degradations across different task categories, as vividly illustrated by the relative performance radar chart for a representative subset of the evaluated models (see Fig. 1). To deeply investigate this dynamic, we formulate four key research questions (RQs) and design corresponding experiments. Furthermore, we introduce two novel metrics—Reasoning-Alignment (RA) and Answer-Alignment (AL)—to quantitatively measure the quality of intermediate images generated during the GtA stage. Through this comprehensive evaluation, we distill our observations into three core findings:

First, we note an overall performance degradation. Integrating generative capabilities does not yield universal gains; rather, unified models generally underperform their corresponding base VLMs on standard understanding tasks. Furthermore, the GtA paradigm—generating intermediate images prior to answering—typically degrades performance relative to direct inference, often propagating visual errors in structurally constrained domains.

Second, despite this general decline, structured improvements consistently emerge. In specific spatial, illusion-sensitive, and multi-round reasoning subtasks, externalizing visual transformations transitions into a significant advantage. In these transformation-intensive scenarios, enhanced spatial and shape perception, as well as multi-step intermediate image states, prove highly beneficial.

Third, we identify a structured task-model correlation. At the task level, G2U gains are not random but cluster by cognitive demands: perception-oriented tasks correlate strongly with one another, as do logic- and reasoning-intensive tasks. At the model level, unified models built upon the same base model demonstrate strong behavioral correlations, whereas those sharing only architectural similarities exhibit much weaker correlations. These highly predictable trends suggest that generation-understanding coupling induces class-consistent inductive biases, which are fundamentally shaped by the underlying pretraining data and, to a lesser extent, the model architectures.

Our main contributions are summarized as follows:

1. **A Novel Testbed:** We introduce UniG2U, the largest unified-model evaluation testbed to date designed specifically for the Generation-to-Understanding (G2U) paradigm. It covers 3,000 carefully curated instances across seven reasoning categories, providing standardized data, protocols, and scripts for reproducible evaluation.
2. **Comprehensive Experiments:** We conduct the largest large-scale study of unified models so far, benchmarking over 30 models across autoregressive, diffusion-based, and hybrid architectures. We systematically isolate G2U gains by strictly comparing unified models against their foundational VLM baselines.
3. **Deep Mechanistic Insights:** Our analyses demystify when generation helps or harms understanding. We expose the inherent architectural trade-offs between generative and discriminative objectives, the vulnerabilities of the Generate-then-Answer visual reasoning paradigm, and the critical role of intermediate alignment fidelity.

2 RELATED WORKS

2.1 UNIFIED MULTIMODAL MODELS

Early multimodal systems largely followed a *disjoint* design, coupling a vision-language understanding model with a separate image generator in a pipeline, which made end-to-end credit assignment across modalities difficult (Alayrac et al., 2022; Li et al., 2023; Yu et al., 2025; Hu et al., 2025a). The rise of large multimodal LMs strengthened the trend toward a *single interaction interface*—using language to control multimodal perception and reasoning—yet image generation often remained externalized as a separate module (Huang et al., 2023). In parallel, diffusion-style text-to-image synthesis made “understanding+generation” pipelines practical at scale (Rombach et al., 2022). More recently, research has pushed toward *unified multimodal models* that can both *understand* and *generate* within one interface, motivated by shared representation learning and tighter coupling between perception and synthesis (Chameleon Team et al., 2024; Xie et al., 2024). Token-based early-fusion models suggest that a single next-token objective over mixed-modal sequences can support both behaviors (Chameleon Team et al., 2024), while hybrid formulations combine autoregressive decoding with discrete diffusion/flow-style generation inside a single transformer to broaden modality patterns (Xie et al., 2024). Scaling these designs toward broader modality coverage (e.g., image/video) has made unified interfaces increasingly competitive as general-purpose multimodal backbones (Xie et al., 2025a; Deng et al., 2025; Chen et al., 2025; Zhang et al., 2025b). Surveys have begun to systematize this design space and repeatedly emphasize that *evaluation* remains a key bottleneck for understanding when unification yields *genuine* capability gains (Yang et al., 2025b; Zhao et al., 2025).

2.2 BENCHMARKS FOR UNIFIED MODELS

Benchmarks have begun to operationalize “unification” in different ways, including capability coverage, mixed-modality output competence, and explicit bidirectional synergy protocols (Xie et al., 2025b; Shi et al., 2025; Zou et al., 2025; Li et al., 2026; 2025b). MME-Unify (MME-U) provides a scalable audit via three domains (understanding, generation, and mixed/unified tasks) and makes unified tasks *quantifiable* by casting them into *text-option* + *image-option* selection,

judging generated images through CLIP-style similarity (Xie et al., 2025b; Radford et al., 2021). RealUnify defines unification as *bidirectional synergy* (UEG/GEU) and introduces paired *direct* and *stepwise* protocols to localize whether failures stem from core skills or integration (Shi et al., 2025). Uni-MMMU centers multi-discipline reasoning with Gen→Und / Und→Gen paradigms and emphasizes verifiable intermediates with dual-channel (text+image) scoring (Zou et al., 2025; Yue et al., 2023). ROVER targets reciprocal cross-modal reasoning, introducing a specific split (ROVER-TG) to evaluate visually-augmented verbal generation, where models produce intermediate visual rationales before answering (Liang et al., 2025). UEval evaluates real-world requests requiring both text and images with item-level rubrics and a runnable pipeline/leaderboard (Li et al., 2026; 2025c).

Despite these advances, existing benchmarks still fall short of diagnosing *generation-to-understanding* (G2U) as a capability phenomenon: (i) what constitutes a *G2U task* where intermediate visuals are intrinsically useful under matched budgets, and (ii) how results vary across fine-grained *task regimes* and *model archetypes*. MME-U offers scale but does not isolate synergy or define G2U task structure (Xie et al., 2025b). RealUnify and Uni-MMMU introduce synergy-aware protocols but cover a narrower slice of task types and unified-model designs (Shi et al., 2025; Zou et al., 2025). While ROVER explores interleaved visual reasoning, its G2U evaluation is limited in scale (404 tasks across 6 subtasks) and does not explicitly pair unified models with their base VLMs to strictly isolate the synergistic gains. UEval focuses on mixed-output quality rather than causal G2U effects (Li et al., 2026).

To bridge this critical gap, we introduce **UniG2U**. As explicitly compared in Table 1, UniG2U stands as the most comprehensive benchmark to date for evaluating unified models. It features the largest sample volume (3,000 understanding tasks), the most diverse taxonomy (30 fine-grained subtasks across 7 reasoning regimes), and the widest model coverage (evaluating over 30 models). More importantly, UniG2U is the first benchmark to systematically and quantitatively analyze the synergistic effect of generation on understanding. By curating G2U-centric regimes (e.g., spatial manipulation, state externalization, visual illusions) and strictly pairing unified models with their purely discriminative base VLMs, we cleanly isolate the G2U capability. Through this unprecedented scale and rigorous framework, UniG2U provides key mechanistic insights into the generation-understanding coupling, revealing class-consistent inductive biases and identifying specific scenarios where generation effectively serves multimodal reasoning.

Table 1: Comparison of unified multimodal benchmarks.

Benchmark	# Understanding Samples	# Subtasks	Synergy
MME-Unify	1964	3	✗
Uni-MMMU	524	4	✓
ROVER	404	6	✓
RealUnify	400	4	✓
UniG2U (ours)	3000	30	✓

3 FORMULATIONS AND DEFINITIONS

This section formalizes the core objects studied in UniG2U. We (i) define *unified multimodal models* (UMMs) and the capability-level notion of *Generation Helps Understanding* (G2U), (ii) introduce a taxonomy of UMMs used throughout the paper, and (iii) distinguish **Direct** vs. **Generate-then-Answer (GtA)** inference under a unified evaluation.

3.1 DEFINITIONS: UMMs AND G2U

Task setting. Each instance is formulated as a tuple $x = (V, T, Q)$, where V denotes the input visual context (e.g., images), T represents the text instructions or context, and Q is the specific query to be answered. The target answer is denoted as $y \in \mathcal{Y}$.

Pure understanding models. A *pure understanding* model maps the given input directly to an answer,

$$\hat{y} = M_U(x), \quad M_U \in \mathcal{M}_U, \quad (1)$$

and does not expose an interface for producing intermediate *visual* artifacts intended to be re-consumed for the same instance.

Unified multimodal models (UMMs). A *unified multimodal model* (M_{UM}) supports multimodal understanding together with a generative interface that can produce an intermediate visual artifact G (e.g., image, sketch, annotation, reconstruction, edit) conditioned on the multimodal input x . Conceptually, UMMs are motivated by the hypothesis that generation can (i) shape internal multimodal representations through unified training, and/or (ii) act as a controllable reasoning resource via these intermediate artifacts.

Defining “Generation Helps Understanding” (G2U). Let $\mathcal{A}(\cdot, \cdot)$ be the task evaluation metric. We define the performance of a model M on a dataset $\mathcal{D}$ as:

$$\text{Perf}(M; \mathcal{D}) = \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{A}(M(x), y)]. \quad (2)$$

To rigorously isolate the effect of generative capabilities, we must mitigate confounders stemming from different architectural designs or pretraining data. Therefore, for a given unified system M_{UM} , let $\mathcal{B}(M_{UM}) \in \mathcal{M}_U$ denote its corresponding *purely discriminative base VLM* (i.e., the foundational understanding model upon which the unified model is built, evaluated under a matched computation budget).

We define the **G2U gain** for the unified system M_{UM} as:

$$\Delta_{G2U}(M_{UM}; \mathcal{D}) = \text{Perf}(M_{UM}; \mathcal{D}) - \text{Perf}(\mathcal{B}(M_{UM}); \mathcal{D}). \quad (3)$$

By enforcing strict **capability comparability** (base-unified pairing) and **budget matching**, we ensure that any observed Δ_{G2U} intrinsically reflects the synergistic effect of generation-to-understanding, rather than mere variations in base model scale or empirical noise.

3.2 TAXONOMY OF UNIFIED MULTIMODAL MODELS

We group evaluated UMMs into three categories based on their architectural designs and training paradigms: **end-to-end unified models** (E2E), **decoupled unified systems** (Decoupled), and **agentic unified models** (UM-Ag).

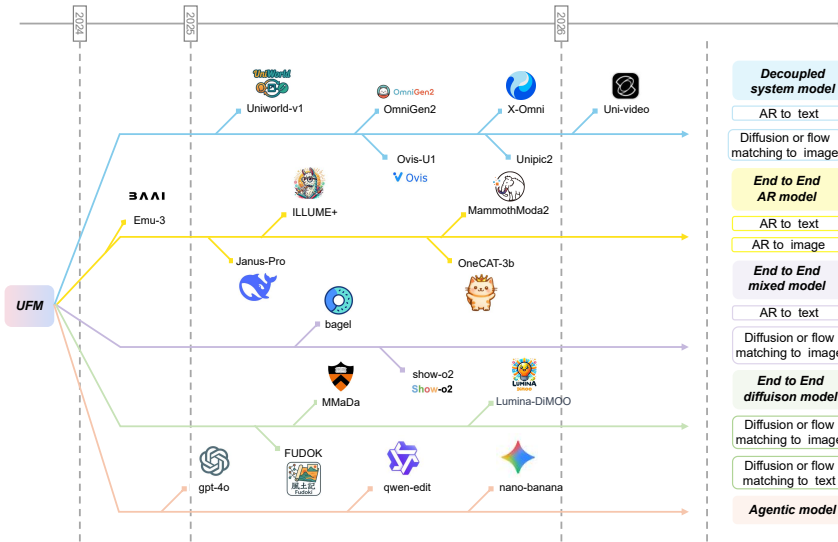


Figure 2: Taxonomy of unified multimodal models (UMMs). All models annotated in the figure are benchmarked in this work.

All categories can be conceptually expressed via a shared factorization:

$$G = \text{Gen}(x), \quad \hat{y} = \text{Ans}(x, G), \quad (4)$$

where $G = \emptyset$ recovers direct answering without intermediate visual artifacts. The taxonomy fundamentally differs in how Gen and Ans are implemented and trained.

(1) End-to-end unified models (E2E). E2E models are *parameter-coupled*: Gen and Ans are realized by a single model with shared parameters θ . Crucially, the generation and understanding capabilities are **jointly trained** (e.g., via interleaved multimodal data) within a unified parameter space. A defining property is that both capabilities emerge from the end-to-end training of one unified architecture, rather than from separated components. Representative instances in our study include *Bagel* and *UniPic2-Metaquery-9B*.

(2) Decoupled unified systems (Decoupled). Decoupled systems are *module-separated*. In contrast to E2E models, the generation and understanding components are **not jointly trained**. Instead, Gen and Ans are implemented by independent modules (with parameters ϕ and ψ) that are stitched together through modular orchestration. Concretely, a generation module produces an explicit intermediate artifact G , which is subsequently consumed by a frozen answering module. This modularity enables flexible component-level splicing. Typical examples include *Bagel two-stage* and *OneCAT*.

(3) Agentic unified models (UM-Ag). In agentic systems, generation and understanding are performed by **completely different models**. They achieve unification merely at the protocol level (via a control policy or bounded tool-use interaction) rather than through architectural integration or joint training. For instance, a pure language model might invoke an external, black-box image generation API. In our study, tool-driven visual exploration settings such as *GPT-4o + image tool* instantiate this category.

Conceptually, we regard UM-Ag as a generalized variant of UMMs. However, in our experimental analysis, we explicitly distinguish agentic models from standard unified models (E2E and Decoupled). This separation is necessitated by the fact that most agentic systems consist of distinct, often closed-source proprietary models that rely on external tool-use rather than undergoing any form of multimodal unified training. Consequently, they inherently lack a strictly matched base VLM, rendering them incompatible with the rigorous base-unified comparative framework defined in Section 3.1 for calculating exact G2U gains.

Fig. 2 summarizes this taxonomy and highlights how the three classes differ in their degree of unification—ranging from jointly trained parameters (E2E) to stitched modules (Decoupled), and finally to tool-invoked separate models (UM-Ag).

3.3 DIRECT VS. GENERATE-THEN-ANSWER (GTA) INFERENCE

The Generation-to-Understanding (G2U) paradigm can be instantiated through two distinct inference protocols: Direct and Generate-then-Answer (GtA). In the *Direct* setting, the system answers the query without producing any external intermediate visual artifacts, formalized as $\hat{y} = \text{Ans}(x, \emptyset)$. In the *GtA* setting, the system explicitly generates an intermediate visual rationale G and re-consumes it to derive the final answer, as defined in (4) (or via an agentic rollout with multiple G_t).

Decomposing G2U gains. Define the best GtA performance for model M as:

$$\text{Perf}_{\text{GtA}}(M; \mathcal{D}) \triangleq \max \left\{ \text{Perf}_{\text{1shot_GtA}}(M; \mathcal{D}), \text{Perf}_{\text{agentic_GtA}}(M; \mathcal{D}) \right\}, \quad (5)$$

where $\text{Perf}_{\text{1shot_GtA}}$ instantiates (4) with a single generated artifact G , and $\text{Perf}_{\text{agentic_GtA}}$ instantiates it with a closed-loop rollout.

We can then decompose the overall G2U performance shift into two specific components:

$$\begin{aligned} \Delta_{\text{Direct}} &= \text{Perf}_{\text{Direct}}(M_{\text{UM}}; \mathcal{D}) - \text{Perf}(\mathcal{B}(M_{\text{UM}}); \mathcal{D}), \\ \Delta_{\text{GtA}} &= \text{Perf}_{\text{GtA}}(M_{\text{UM}}; \mathcal{D}) - \text{Perf}_{\text{Direct}}(M_{\text{UM}}; \mathcal{D}). \end{aligned} \quad (6)$$

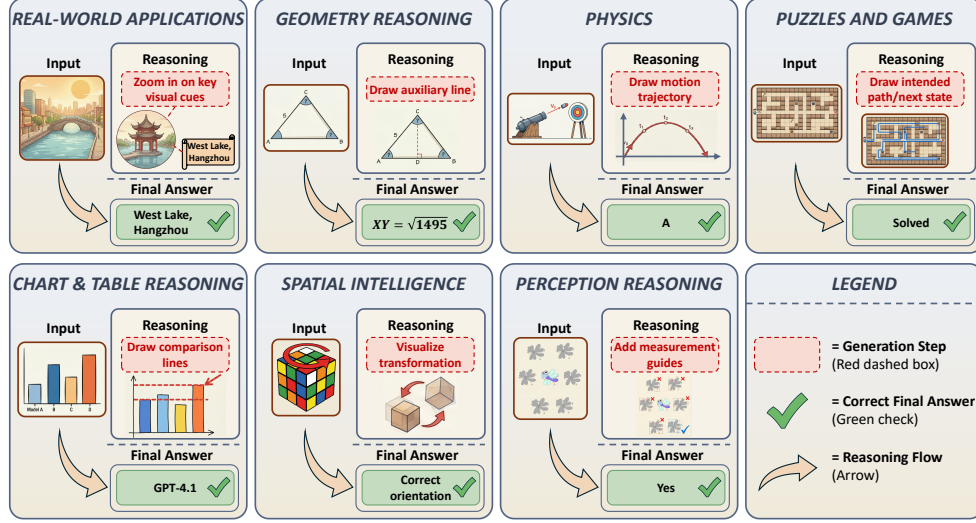


Figure 3: Task taxonomy overview of UniG2U.

Here, Δ_{Direct} captures the intrinsic capability shift resulting from unified multimodal training (without explicit generation), while Δ_{GtA} isolates the specific performance impact of explicitly externalizing the visual reasoning process prior to answering.

While externalizing visual transformations can be beneficial, the GtA protocol can also degrade performance if the intermediate generated artifacts are inaccurate, physically implausible, or misaligned with the intended reasoning plan, thereby misleading the subsequent answering module.

4 THE SETUP OF UNIG2U

4.1 DATASET CONSTRUCTION AND TASK TAXONOMY

UniG2U is designed as a diagnostic benchmark to study when and why generation benefits multimodal understanding. Rather than focusing on a single task family, we construct a diverse dataset covering a wide spectrum of perceptual and cognitive structures. In total, UniG2U contains approximately **3,000 samples**, organized into **7 high-level categories** and **30 fine-grained subcategories**. This section details the taxonomy design and the motivation behind each category.

4.1.1 DESIGN PRINCIPLES

The dataset taxonomy is guided by two core principles, fundamentally echoing Richard Feynman’s profound adage: “*What I cannot create, I do not understand.*”

First, categories should reflect *distinct underlying cognitive structures* rather than superficial differences in dataset sources.

Second, all selected categories must theoretically exhibit the potential for Generation-to-Understanding (G2U) synergy. Rather than aggregating arbitrary multimodal tasks, we intentionally curate scenarios where visual externalization is intrinsically coupled with comprehension. Whether through drawing geometric auxiliary lines, simulating physical dynamics, tracking spatial transformations, or even reconstructing fine-grained perceptual features.

1. **Real-world Applications (200 samples).** Visually grounded tasks resembling practical applications, such as attentional focusing and visual shortest-path reasoning (VSP). These tasks typically require identifying relevant regions, tracking spatial relations, or following

visual constraints across complex scenes. Data are primarily sourced from VSP-style benchmarks and attentional reasoning datasets.

2. **Geometry Reasoning (200 samples).** Visual mathematics and geometry problems, including both 2D and 3D geometric reasoning. Subtasks include planar and solid geometry, curated from datasets such as Geometry3K and AuxSolidMath. These problems often benefit from structured intermediates (e.g., drawing auxiliary lines, reconstructing geometric structure), making them ideal for evaluating explicit generation-augmented understanding.
3. **Physics Reasoning (200 samples).** Known-answer scientific reasoning grounded in diagrams, including physics and optics problems. We include curated subsets from datasets such as Clouddriver and PhyX, where solving the problem requires interpreting diagrams, understanding physical processes, or reasoning over scientific illustrations.
4. **Puzzles and Games (537 samples).** Structured reasoning with state tracking and rule-based transformations, including mental reconstruction/tracking, visual tracking, and classical puzzles such as maze navigation, jigsaw puzzles, and sliding puzzles. Data are drawn from multiple sources including Uni-MMMU, RealUnify, and BabyVision-style datasets. These tasks often admit explicit intermediate visual states, making them suitable for studying both implicit and explicit generation mechanisms.
5. **Chart & Table Reasoning (100 samples).** Reasoning over structured visual data (charts/tables). We use a curated subset of ChartQA, where models must interpret bar charts, line plots, and other visualized representations; intermediate generation may help by re-expressing or simplifying the chart.
6. **Spatial Intelligence (500 samples).** Reasoning about spatial relations, geometric transformations, and object dynamics (e.g., multi-step spatial reasoning, attribute measurement/approximation, camera/object motion). Samples are drawn primarily from spatial reasoning datasets such as MMSI-Bench.
7. **Perception Reasoning (1263 samples).** Fine-grained visual perception and low-level reasoning, including scene/shape-based reasoning, in-scene vs. in-shape discrimination, logo understanding, algorithmic and deductive visual reasoning, spatial pattern recognition, and analogical/inductive reasoning. Data are curated from IllusionBench, visual puzzle datasets, and BabyVision-style benchmarks; compared to other categories, many tasks here are primarily perceptual and may benefit less from explicit generation.

By structuring the benchmark across seven categories with distinct reasoning characteristics, UniG2U enables fine-grained analysis of the relationship between generation and understanding. As illustrated in Fig. 3, the taxonomy spans both generation-friendly categories (e.g., spatial reasoning, puzzles) and more perception-heavy categories, allowing us to probe when intermediate visual generation is helpful, neutral, or harmful. Instead of merely reporting aggregate accuracy, this design allows us to investigate how the effectiveness of generation varies across cognitive regimes, providing deeper insight into the mechanisms of unified multimodal models.

4.1.2 TASK SELECTION AND COVERAGE

UniG2U is constructed around a single guiding hypothesis: there exists a family of multimodal understanding tasks for which intermediate visual generation can serve as an effective reasoning aid. Such tasks are common in practice, including (i) **spatial reasoning** where models benefit from externalizing transformations or layouts, (ii) **chart & table reasoning** where re-expression or simplification of plots can reduce cognitive load, (iii) **puzzles and games** where intermediate states can be visualized to support state tracking, and (iv) **Physics/geometry** problems where auxiliary constructions and diagram manipulation often enable correct solutions.

To ensure both *comprehensive coverage* and *strong relevance* to the G2U objective, we curate **3,000** instances across **7** high-level categories and **30** subtasks (Table 2). As shown in Fig. 4, our candidate pool is collected from diverse public sources (e.g., IllusionBench, MMSI-Bench, Visual Puzzles, Geometry3K, AuxSolidMath, Uni-MMMU, ChartQA, RealUnify, BabyVision, PhyX, and VSP-style benchmarks), from which we further *select and filter* instances that best reflect settings where intermediate generation is likely to assist understanding.

Table 2: **Overview of the UniG2U dataset.** The benchmark contains 3,000 samples organized into 7 reasoning categories and 30 fine-grained subtasks, thoughtfully curated from multiple established multimodal reasoning datasets.

Category	Subtasks	Data Sources	# Samples
Real-world Applications	Attentional Focusing; Visual Shortest-Path	VSP Li et al. (2025a), RealUnify Shi et al. (2025)	200
Geometry Reasoning	Planar Geometry; Solid Geometry	Geometry3K Lu et al. (2023), AuxSolidMath-Easy Guo et al. (2026)	200
Physics Reasoning	Mechanics; Optics	PhyX Zhang et al. (2025a)	200
Puzzles and Games	Mental Reconstruction; Mental Tracking; Visual Tracking; Maze; Jigsaw; Sliding	Uni-MMMU Zou et al. (2025), RealUnify Shi et al. (2025), BabyVision Chen et al. (2026)	537
Chart & Table Reasoning	Chart Question Answering	ChartQA Masry et al. (2022)	100
Spatial Intelligence	MSR; Attribute (Measurement); Attribute (Approximation); Motion (Camera); Motion (Object)	MMSI-Bench Yang et al. (2025a)	500
Perception Reasoning	Icon-Scene; Icon-Shape; In-scene; In-shape; Logo-Scene; Logo-Shape; Algorithmic; Deductive; Spatial; Analogical; Inductive	IllusionBench Zhang et al. (2025d), Visual Puzzles Wang et al. (2025a), BabyVision Chen et al. (2026)	1263
Total	30 fine-grained subtasks	Multiple sources	3,000

In particular, the benchmark includes 200 samples each for Real-world Applications, Geometry Reasoning, and Physics Reasoning; 537 samples for Puzzles and Games; 100 samples for Chart & Table Reasoning; 500 samples for Spatial Intelligence; and 1,263 samples for Perception Reasoning. This distribution allows us to analyze generation benefits across cognitive regimes ranging from generation-friendly spatial manipulation to primarily perceptual recognition.

It is worth noting that while the sample size per fine-grained subtask may appear relatively modest, this is a deliberate consequence of our rigorous selection criteria. Tasks that genuinely exhibit G2U synergy—where intermediate visual externalization provides a tangible reasoning advantage—are intrinsically scarce in existing datasets, which are historically skewed towards direct perception or text-only reasoning. Despite this inherent scarcity, UniG2U’s collection of 3,000 meticulously curated samples renders it the largest benchmark of its kind compared to existing unified evaluations (as quantitatively compared in Table 1). Coupled with our extensive evaluation across more than 30 distinct unified models and baselines, UniG2U represents a truly large-scale, comprehensive, and highly persuasive investigation into the multimodal unification landscape.

Each instance in UniG2U is organized around a *question-answering* format with an image (or a small set of images) as the primary input. In addition to the final answer, we annotate (when applicable) auxiliary information to facilitate diagnostic evaluation:

- **Category / Subtask labels** for fine-grained analysis across the taxonomy.
- **Answer type** (e.g., multiple-choice, short text, numeric), which determines post-processing and matching rules.
- **Generation requirement tags** indicating whether a task naturally admits helpful intermediate visual representations (e.g., geometric construction, state transition visualization, chart re-expression).

Importantly, UniG2U does *not* force a single canonical intermediate image for every problem. Instead, we treat intermediate generation as a *reasoning tool* whose usefulness should be measured by whether it improves final task performance.

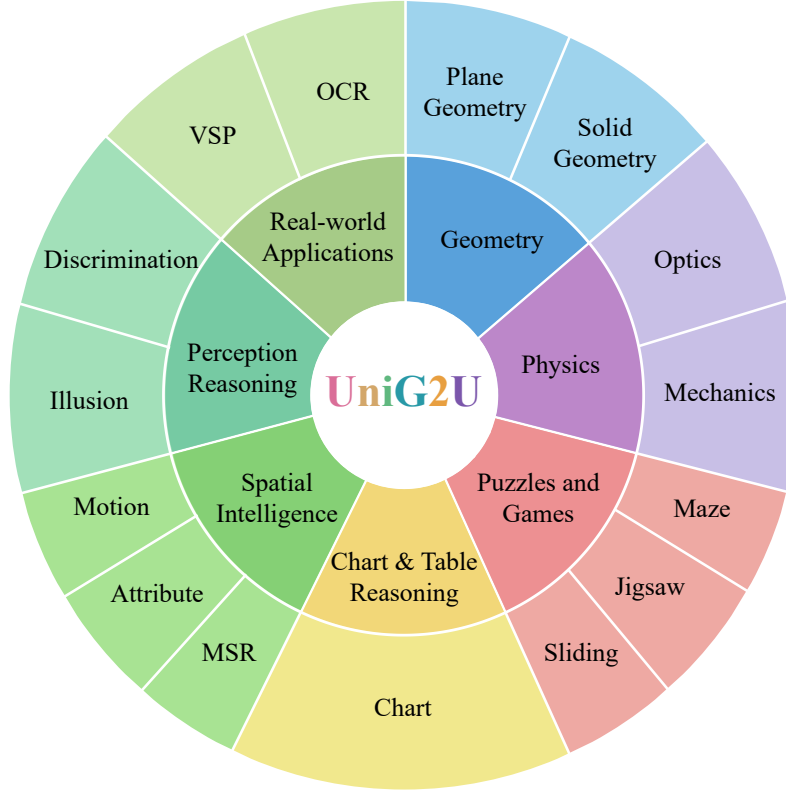


Figure 4: Task selection and coverage in UniG2U.

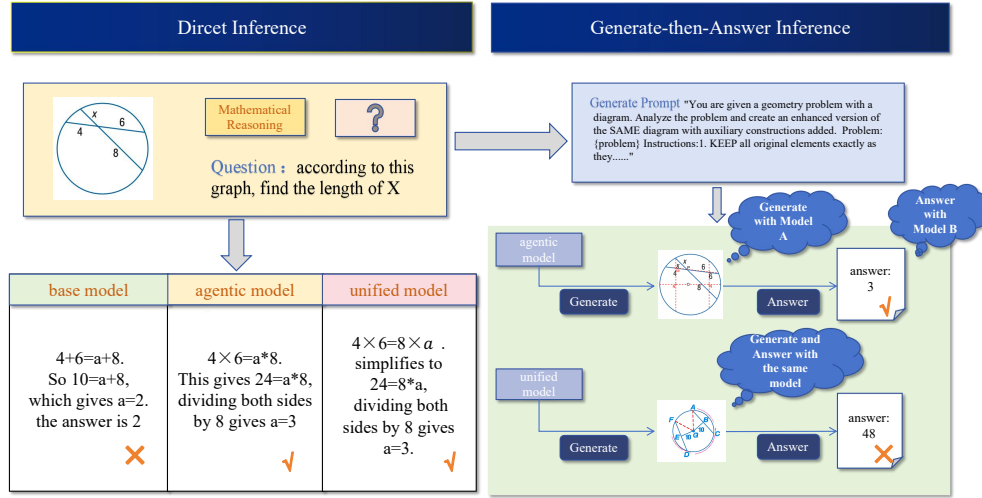


Figure 5: Overview of our two evaluation prompt formats: Direct and Generate-then-Answer (GtA).

4.2 EVALUATION PROTOCOL

Our evaluation protocol is strictly designed to ensure *fair comparison* across a diverse suite of models and to rigorously isolate the *causal contribution of intermediate generation* (i.e., the G2U effect).

Unified evaluation environment. To minimize implementation biases, we execute all evaluations under a unified architecture using the `lmms-eval` framework whenever possible. For proprietary models or specific architectures incompatible with this environment, we fallback to their native evaluation stacks while strictly preserving the rest of our protocol (i.e., prompt templates, budget constraints, and decoding strategies).

Dual-protocol prompting (Direct vs. GtA). To localize the effect of generation, we evaluate all models under two paired inference protocols, as illustrated in Fig. 5:

- **Direct:** The model processes the task instruction, input image(s), and query to directly output the final answer in a single end-to-end forward pass.
- **Generate-then-Answer (GtA):** The instance is explicitly decomposed. The model is first prompted to generate an intermediate visual artifact G (e.g., drawing auxiliary lines or tracking states). Subsequently, it generates the final answer using both the original input and the newly generated G as expanded context.

For inherently multi-step reasoning problems (e.g., maze navigation, jigsaw puzzles), we employ an *iterative GtA* variant: if an instance requires K steps, the model alternates between producing a step- k intermediate visual state and the corresponding step- k decision, until the final answer is reached. Models unable to support the GtA mode due to architectural constraints receive specific fallback treatments, detailed in Appendix B.

Budget matching and interaction constraints. To prevent performance gains from being confounded by disparate compute budgets, we enforce strict bounds. Across all models and task categories, we cap the maximum token generation length. Input images are processed at the model’s default resolution; if out-of-memory (OOM) exceptions occur, we downscale to the nearest supported resolution.

Unified decoding settings. To ensure deterministic and reproducible evaluations, we standardize the text decoding hyperparameters across all models using greedy decoding (temperature = 0, top- p = 1.0). For intermediate image generation, we generally adopt the default configurations of the respective models.

4.3 METRICS

Accuracy. We report accuracy (or exact match) as the primary baseline metric. Depending on the task format, we compute correctness via (i) *rule-based evaluation* using text normalization followed by regex matching, or (ii) *LLM-as-a-judge evaluation* by invoking the GPT-4o API with task-specific grading prompts. Unless otherwise noted, GPT-4o is strictly used to adjudicate non-trivial free-form outputs and visual-dependent judgments.

G2U Gain. Our core metric is the **G2U Gain** (Δ), defined as the accuracy improvement of a unified model M_{UM} relative to its rigorously paired purely discriminative base model $\mathcal{B}(M_{UM})$. Recalling Eq. 6, we report two distinct variants:

$$\Delta_{\text{Direct}} = \text{Acc}_{M_{UM}}^{\text{Direct}} - \text{Acc}_{\mathcal{B}(M_{UM})}, \quad (7)$$

$$\Delta_{\text{GtA}} = \text{Acc}_{M_{UM}}^{\text{GtA}} - \text{Acc}_{\mathcal{B}(M_{UM})}. \quad (8)$$

This relative formulation effectively isolates the net capability shift introduced by the unified architecture and the generation process. We compute the G2U Gain both overall and per subtask to quantitatively diagnose when generation helps, remains neutral, or harms multimodal understanding.

Diagnostic metrics for GtA intermediates: RA and AL. To mechanically understand the effectiveness (or failure) of intermediate images produced during the GtA protocol, we introduce two novel alignment metrics evaluated by GPT-4o on a 1-to-5 scale, using customized rubrics for each task family. For SenseNova-U1.5-8B-MoT, the supplied RA/AL scores were produced by GPT-5.5 with the same task-specific rubrics; this evaluator difference is marked in Table 7.

RA METRIC

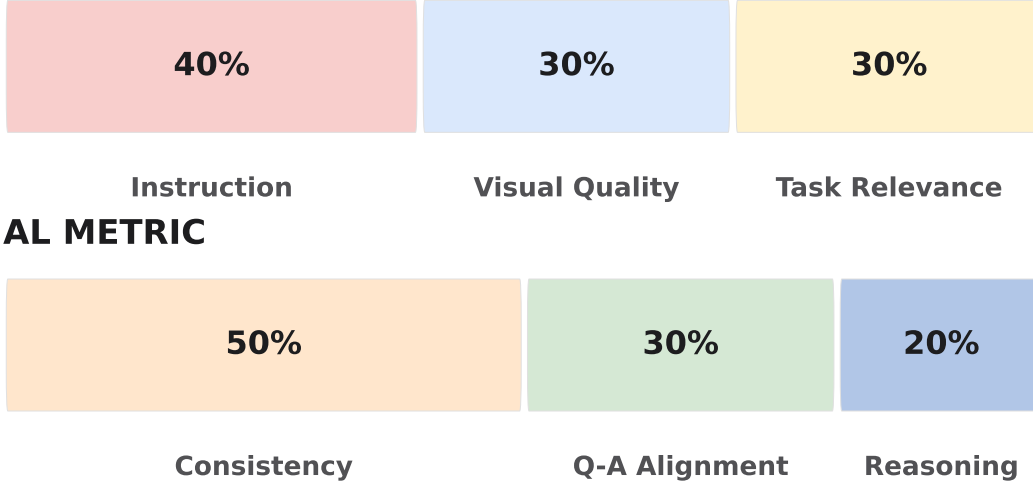


Figure 6: Rubric weights for GtA intermediate evaluation. RA measures the adherence and fidelity of the generated intermediate image w.r.t. the generation prompt, while AL measures the logical consistency between the final answer, the generated image, and the original question.

Reasoning-to-Visual Alignment (RA). RA evaluates whether the generated intermediate image correctly materializes the intended reasoning scaffold. It aggregates instruction adherence, visual quality, and task relevance (weighting schema detailed in Fig. 6). Essentially, RA measures the *fidelity of visual externalization*.

Answer-to-Visual Alignment (AL). AL assesses whether the model’s final answer is logically consistent with both the generated intermediate image and the original query. It aggregates visual-answer consistency, question-answer alignment, and reasoning coherence (see Fig. 6). AL effectively measures the *re-consumption capability* of the generated visual context.

Scoring schema. For both RA and AL, the evaluator produces a final score on a 5-point scale by aggregating rubric-specific sub-scores. We report the mean RA/AL scores across relevant instances, providing granular breakdowns for each model and subtask to correlate intermediate generation quality with ultimate task accuracy.

4.4 MODELS AND IMPLEMENTATION DETAILS

To comprehensively study the G2U effect under realistic model ecosystems, we systematically evaluate a massive suite of **36 distinct models**. These are categorized into three families: **11** base VLMs, **22** native unified models, and **3** agentic models.

Base VLMs (understanding-only). We benchmark 11 strong vision-language models (VLMs) that perform *pure understanding* without generating intermediate images. This set comprises the exact corresponding base models for all tested unified models, alongside leading VLM models GPT-4o and Ovis2.5. These base VLMs serve as the strict reference points for computing the G2U Gain (Equations (7) and (8)).

Native unified models. We benchmark 22 representative open-weight unified models that natively integrate multimodal understanding and image generation within a single model interface (i.e., the model can directly follow the **Direct** and **GtA** protocols without external tool wiring). These models serve as our primary subjects for evaluating intrinsic understanding-through-generation. Crucially, as detailed in Table 3, the **Backbone** column denotes the corresponding Base VLM when a paired baseline is available; SenseNova-U1.5 has no paired base result in the supplied evaluation snapshot.

Table 3: **Comparison of Unified Multimodal Models.** * indicates frozen backbone,  indicates unfrozen (trainable) backbone. The **Backbone** column represents the paired Base VLM when one is available. SenseNova-U1.5 has no paired base score; OPD denotes multi-expert on-policy distillation.

Model	Date	Backbone	Modality	Image Paradigm	Training	i2i	Unify	Text Paradigm
Emu3	2024.09	LLaMA-2 	image, audio	diffusion	UP+ST+AT	no (OOD)	end-to-end	AR
X-Omni	2025.07	Qwen-2.5 	image	AR	UP+ST+AT	no (OOD)	decoupled	AR
UniWorld-V1	2025.06	Qwen-2.5-VL *	image	diffusion	UP+ST	yes	decoupled	AR
MIO	2025.10	Yi-6B-Base 	image, audio	diffusion	UP+ST	yes	decoupled	AR
Bagel	2025.05	Qwen-2.5 	image, video	Flow	UP+ST	yes	end-to-end	AR
OmniGen2	2025.06	Qwen-2.5-VL 	image	Flow	UP+ST	yes	decoupled	AR
Ovis-U1	2025.06	Qwen3-1.7B 	image	diffusion	UP+ST	yes	decoupled	AR
OneCAT-3B	2025.09	Qwen-2.5-VL 	image	AR	UP+ST	yes	end-to-end	AR
JavisGPT	2025.12	Qwen2.5-VL 	video	diffusion	UP+ST	no	decoupled	AR
UAE	2025.10	Qwen-2.5-VL-3B 	image	diffusion	UP+ST+AT	no (OOD)	decoupled	AR
TokenFlow-XL	2024.12	Qwen-2.5-14B 	image	AR	UP+ST	no (OOD)	end-to-end	AR
UniPic2	2025.09	Qwen-2.5-VL *	image	diffusion	UP+ST+AT	yes	decoupled	AR
Janus-Pro	2025.01	DeepSeek-LLM 	image	AR	UP+ST	no (OOD)	end-to-end	AR
Show-o2	2025.06	Qwen2.5-7B 	image, video	Flow	UP+ST	yes	end-to-end	AR
ILLUME+	2025.04	Qwen-2.5 	image	AR + diffusion	UP+ST	yes	end-to-end	AR
STAR-7B	2025.12	Qwen-2.5-VL *	image	AR + diffusion	UP+ST	yes	end-to-end	AR
MMaDA	2025.05	LLaDA-Instruct 	image	Discrete Diffusion	UP+ST+AT	no (OOD)	end-to-end	Discrete Diffusion
FUDOKI	2025.05	Janus-1.5B 	image	Discrete Flow	UP+ST	no (OOD)	end-to-end	Discrete Flow
AIA	2025.12	Janus-Pro 	image	AR	ST	no (OOD)	end-to-end	AR
Uni-Video	2026.01	Qwen2.5-VL-7B *	image, video	diffusion	UP+ST	yes	decoupled	AR
MammothModa2	2025.11	Qwen3-VL-8B 	image	AR + diffusion	UP+ST+AT	yes	end-to-end	AR
SenseNova-U1.5	2026.09	–	image	patch reconstruction	MoT + OPD	yes	end-to-end	N/A

Agentic models Finally, we evaluate 3 agentic models by composing a planner LLM with an external image generation tool. As established in our taxonomy, agentic systems are conceptually regarded as a generalized variant of unified models. Concretely, the planner produces a `generation_prompt`, invokes the generation API to obtain an intermediate image, and then answers the query conditioned on this newly generated visual state. Notably, these agentic systems are typically driven by the most powerful, closed-source proprietary models available (e.g., **GPT-4o + GPT-image** and **Gemini + Nano Banana Pro**). Evaluating this category serves a critical dual purpose: on one hand, it benchmarks the performance upper-bound (ceiling) of generalized unified models on our tasks; on the other hand, it allows us to investigate how non-end-to-end, purely tool-driven visual externalization performs.

When a unified model does not have an official corresponding VLM base model (or its base model is not a vision-language model), we select the closest available VLM baseline in terms of training data and architecture, and use it as the strictly paired base model for comparison.

Models evaluated in this work. We evaluate all 22 unified models listed in Table 3, including: Emu3 Wang et al. (2024), X-Omni Geng et al. (2025), UniWorld-V1 Lin et al. (2025), MIO Hu et al. (2025b), Bagel Deng et al. (2025), OmniGen2 Xiao et al. (2024), Ovis-U1 Wang et al. (2025b), OneCAT-3B Peng et al. (2025b), JavisGPT Liu et al. (2025), UAE Yuan et al. (2025), TokenFlow-XL Li et al. (2024), UniPic2 Peng et al. (2025a), Janus-Pro Chen et al. (2025), Show-o2 Xie et al. (2025a), ILLUME+ Zhang et al. (2025c), STAR-7B Li et al. (2025d), MMaDA Yang et al. (2025c), FUDOKI Wang et al. (2025c), AIA Zheng et al. (2025), Uni-Video Wei et al. (2025), MammothModa2 Shen et al. (2025), and SenseNova-U1.5-8B-MoT Diaio et al. (2026).

An overview of these models and their key design choices is provided in Table 3, detailing their backbones (Base VLMs), supported modalities, image generation paradigms, training recipes, i2i capability, and whether the unification is end-to-end or decoupled.

All systems are evaluated under the same dataset, prompt templates, and budget constraints described in Section 4.2.

4.5 RESEARCH QUESTIONS

We define four primary research questions (RQs) to systematically guide our empirical study of the Generation-to-Understanding (G2U) paradigm. We systematically address these RQs in the subsequent Results section.

RQ1: Unified vs. Base Models. Do unified multimodal models exhibit consistent capability advantages over their strictly paired base VLMs on our diagnostic benchmark?

Experiment design: For each unified model, we identify its corresponding purely discriminative base VLM and evaluate both under identical datasets, prompt templates, and inference budgets. We report the absolute performance and the Direct G2U Gain (Δ_{Direct}) overall and per cognitive category to characterize where implicit parametric unification yields benefits.

RQ2: Generate-then-Answer (GtA) vs. Direct Inference. For complex reasoning tasks, do unified models benefit more from the *explicit* visual externalization of intermediate steps (GtA protocol), or does *Direct* inference achieve comparable understanding performance?

Experiment design: We evaluate each unified model under both the **Direct** and **GtA** protocols (Section 4.2) using matched compute budgets. We then quantify the performance delta (Δ_{GtA}) between the two modes to strictly isolate the causal contribution of explicitly generating intermediate visual reasoning scaffolds.

RQ3: Behavioral Correlation within Architectural Classes. Within specific architectural families of unified models, do we observe correlated performance patterns that suggest class-consistent strengths and vulnerabilities in G2U tasks?

Experiment design: Following the taxonomy in Table 3, we group models by their image-generation paradigms (e.g., Autoregressive vs. Diffusion vs. Flow) and unification strategies (End-to-End vs. Decoupled). We construct model-by-task performance vectors and compute pairwise rank correlations to determine if fundamental architectural choices dictate distinct G2U behaviors.

RQ4: Fidelity and Utility of Intermediate Visualizations. When explicitly prompted to generate intermediate auxiliary images under the GtA protocol, do unified models reliably follow instructions, and do they produce helpful visuals that genuinely facilitate downstream reasoning?

Experiment design: We diagnose the generated intermediate outputs using the two rubric-based evaluation metrics defined in Section 4.3: **Reasoning-to-Visual Alignment (RA)** and **Answer-to-Visual Alignment (AL)**. We then analyze the correlation between RA/AL scores, downstream task accuracy, and the observed GtA-vs-Direct gains to mechanically trace the source of G2U successes and failures.

5 RESULTS ANALYSIS

5.1 MAIN RESULTS (RQ1)

Table 4 summarizes the performance of all evaluated models across the seven cognitive categories of UniG2U. To rigorously answer RQ1, we group unified models with their strictly paired base VLMs (marked with *). The metric Δ isolates the absolute G2U Gain (or loss) introduced by the unified architecture. SenseNova-U1.5 is reported as an unpaired model: its absolute scores are included, but its Δ is not computed because no matched base result was supplied.¹

Overall Trends: The “Alignment Tax” of Unification.

Takeaway 1: On the majority of tasks, integrating generation capabilities leads to a degradation in pure understanding performance.

One might naturally hypothesize that this degradation in pure understanding is merely a symptom of “catastrophic forgetting” during the multimodal joint training phase. However, empirical evidence solidly refutes this. The native unified models evaluated in our benchmark undergo extensive co-training on massive understanding datasets alongside generation data, a recipe explicitly designed to preserve pure comprehension capabilities. Therefore, rather than a loss of previously acquired

¹**Evaluation constraints:** To ensure strictly fair comparisons, we exclude GtA results for models inherently lacking image-editing training (OOD for visual modification). Similarly, we omit Direct results for models with frozen backbones, as their performance deviations stem from projection layers rather than parameter-level multimodal coupling. Detailed treatments and analyses are deferred to Appendix B and A.

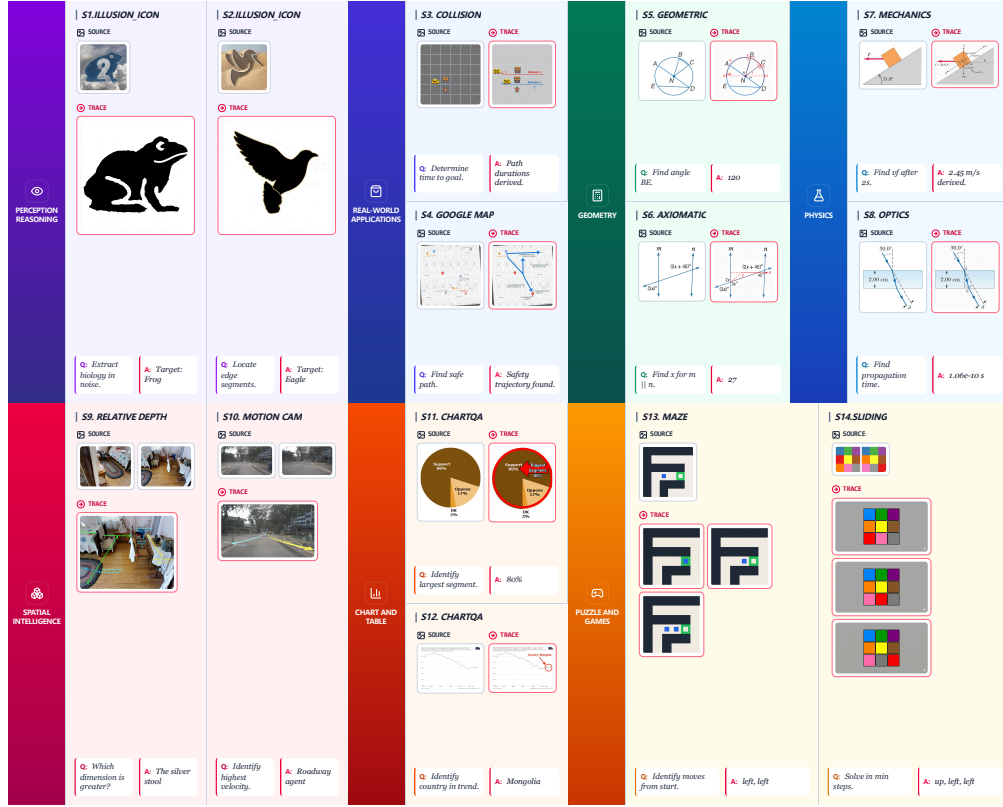


Figure 7: Representative case studies from UniG2U-Bench

knowledge, the observed degradation must inherently stem from the cross-modal “alignment tax” and the objective interference introduced by routing through the generative pathway.

A striking observation across Table 4 is that unified models frequently yield negative Δ values compared to their corresponding base VLMs. Crucially, this degradation persists even under the *Direct* protocol (where no explicit image is generated). This indicates that the decline is not merely a byproduct of low-quality intermediate visual artifacts, but rather an intrinsic consequence of the *parameter-level coupling* between generation and understanding.

This phenomenon suggests a representational trade-off, akin to an “alignment tax.” Joint multimodal training objectives intended to fuse generation and perception often introduce **objective interference**, slightly compromising the fine-grained discriminative reasoning abilities originally present in the base understanding models.

Category-Level Behaviors: Isolated Gains Amidst Overall Decline.

Takeaway 2: Despite the general degradation in overall understanding, unified models exhibit isolated resilience and notable gains in Spatial Intelligence and Visual Illusions.

A finer-grained category analysis reveals that the negative impact of parameter-level coupling is not absolute. While many general reasoning categories suffer from objective interference, unified models consistently demonstrate clearer resilience or even modest gains in **Spatial Intelligence** and **Visual Illusions** (a specific subset of Perception Reasoning).

To illustrate this trend, Table 5 presents an excerpt of the results focusing on the `llava_onevision` model family. As shown, despite the overall “alignment tax,” generation-integrated models like `bagel` and `show-o2` achieve notable positive gains (Δ) in tasks requiring transformation-aware reasoning and geometric consistency. For the complete experimental results across all evaluated base VLMs, as well as a more detailed analysis of how generation-induced representations regularize internal spatial structures, please refer to Appendix C.

Table 4: **Main results on UniG2U across categories.** All values are reported as percentages ($\times 100$). The best result within each model group is **bolded**. Baseline models (marked with *) are highlighted with gray background. Δ denotes the performance gap to the respective baseline.

Model	Real-world Apps	Geometry Reasoning	Physics	Puzzles & Games	Chart	Spatial Intel.	Perception Reasoning	Overall (G2U Gain Δ)
qwen2.5-v1-7b*	40.25	23.00	34.00	25.73	86.00	23.40	39.42	34.45 (+0.00)
OmniGen2(direct)	41.25	14.00	41.00	20.91	86.00	23.80	35.62	31.99 (-2.46)
OmniGen2(GtA)	41.25	13.00	41.50	21.91	79.00	26.40	34.52	31.87 (-2.58)
UniWorld-V1(GtA)	24.00	19.00	34.50	24.72	80.00	24.80	39.35	32.96 (-1.49)
OneCAT-3b(direct)	39.50	17.00	32.00	22.05	76.68	25.20	34.55	31.15 (-3.30)
OneCAT-3b(GtA)	33.50	8.50	29.00	20.69	74.00	23.80	33.10	28.80 (-5.65)
JavisGPT(direct)	32.00	21.00	26.50	23.72	78.60	21.60	35.52	30.72 (-3.73)
uni-video(GtA)	34.00	21.50	29.00	29.33	61.00	20.00	42.76	34.25 (-0.20)
UniPic2(GtA)	17.50	17.00	31.00	23.95	47.00	24.60	38.80	30.65 (-3.80)
STAR-7B(GtA)	34.00	14.50	32.50	23.56	79.00	25.20	38.56	32.68 (-1.77)
Qwen2.5-VL-3B*	41.00	15.50	40.50	20.42	87.00	26.40	35.55	32.39 (+0.00)
UAE(direct)	36.75	10.50	30.50	19.06	85.00	25.20	36.37	30.94 (-1.45)
Ovis-U1(direct)	45.50	23.50	30.50	27.28	90.00	17.60	34.92	32.15 (-0.24)
Ovis-U1(GtA)	21.00	16.50	17.00	18.00	32.00	20.40	30.56	24.19 (-8.20)
yi-6B-v1*	24.00	0.00	33.50	13.22	19.00	22.60	31.20	23.73 (+0.00)
mio(direct)	18.00	0.00	31.00	12.85	15.00	23.40	25.49	20.70 (-3.03)
mio(GtA)	15.00	0.00	28.00	8.94	13.00	21.60	24.94	19.00 (-4.73)
deepseek-v2*	11.00	4.50	22.00	18.02	74.00	17.40	35.07	25.86 (+0.00)
Janus-Pro(direct)	27.50	7.50	19.00	22.49	29.00	24.20	35.08	27.39 (+1.53)
Ala(direct)	13.50	8.50	31.00	23.95	33.00	26.60	36.34	28.65 (+2.79)
LLaDA-Instruct*	27.25	1.50	21.00	20.83	9.00	23.40	13.78	17.05 (+0.00)
MMaDA(direct)	28.75	5.00	26.50	16.01	14.00	25.00	22.74	21.09 (+4.04)
Janus-1.3B*	13.50	2.50	29.00	12.10	50.00	21.80	23.52	20.36 (+0.00)
FUDOK(direct)	17.50	1.00	21.00	9.31	22.00	29.00	15.52	16.40 (-3.96)
Qwen3-VL-8B*	40.00	39.00	37.00	24.24	89.00	26.00	43.66	37.75 (+0.00)
MammothModa2(direct)	33.50	0.00	36.50	12.10	91.00	21.80	39.11	29.97 (-7.78)
llava-hf*	23.00	0.00	29.50	18.96	28.00	24.60	33.57	26.06 (+0.00)
Emu3(direct)	16.50	1.00	20.50	13.37	67.00	23.20	24.78	21.46 (-4.60)
llava-onevision*	43.00	10.50	44.00	24.48	78.00	25.40	37.13	33.35 (+0.00)
X-Omni(direct)	33.00	9.50	29.50	26.24	83.00	27.00	35.31	31.63 (-1.72)
bagel(direct)	38.00	29.00	37.50	23.77	92.00	28.40	39.96	35.84 (+2.49)
bagel(GtA)	32.75	30.00	40.50	31.58	91.00	29.00	37.29	36.10 (+2.75)
TokenFlow-XL(direct)	32.00	21.00	26.50	23.72	78.60	21.60	35.52	30.72 (-2.63)
show-o2(direct)	35.50	11.00	40.00	26.01	45.00	25.80	36.50	31.59 (-1.76)
show-o2(GtA)	33.00	9.00	40.00	26.39	35.00	20.40	28.11	26.59 (-6.76)
ILLUME+(direct)	24.50	7.50	35.00	21.06	80.00	23.20	35.08	29.54 (-3.81)
ILLUME+(GtA)	20.50	2.00	35.50	19.53	58.00	21.00	34.05	27.13 (-6.22)
gpt4o	45.25	27.00	34.50	30.73	58.00	36.00	43.73	38.96
ovis2.5	44.00	32.50	43.00	23.13	89.00	27.80	42.28	37.51
GPT-4o + GPT-image	32.75	27.00	30.50	32.72	60.00	27.00	40.54	35.44
gemini pro +nano banana pro	71.50	85.00	91.00	55.31	68.00	38.40	61.12	60.80
Qwen2.5-7b+Qwen-edit	33.75	17.00	31.50	27.79	80.00	21.60	38.56	32.96
SenseNova-U1.5-8B-MoT (Direct)	42.50	44.50	32.50	22.72	85.00	29.20	38.82	36.08 (N/A)
SenseNova-U1.5-8B-MoT (Visual CoT)	39.50	45.50	34.00	24.83	82.00	30.00	40.42	37.13 (N/A)

Note: * denotes the base model. Δ is computed as overall minus the base model's overall within each block. SenseNova's Δ is N/A because no paired-base scores were supplied; Direct uses `enable_thinking=false` and Visual CoT corresponds to the workbook's Visual CoT setting. **Bold:** Within each block (same model family), the best score in each category is highlighted in bold; ties are broken by bolding the first occurrence.

These specialized tasks heavily involve geometric transformations, state tracking, and resolving complex visual ambiguities. In these specific domains, the integration of generative capabilities does not act as an interfering distraction; rather, it appears to serve as a powerful structural regularizer. By learning to synthesize and manipulate images, the model significantly enhances its foundational *perceptual acuity* and its ability to explicitly *model visual structures*. This suggests that while current unification paradigms may dilute general reasoning, they hold distinct and unique promise for tasks strictly grounded in structural and spatial modeling.

Table 5: **Excerpt of Absolute Results on Selected Illusion and Spatial Subtasks.** Baseline model is highlighted in gray. See Appendix C for the comprehensive table encompassing all model blocks.

Model	Illusion		Spatial Intelligence			Overall (Δ)
	icon_shape	in_shape	MSR	Attr. (Meas.)	Motion (Cam.)	
llava_onevision*	6	18	17	26	20	17.40 (+0.00)
X-Omni(direct)	5	11	20	31	24	18.20 (+0.80)
bagel(direct)	2	18	26	31	26	20.60 (+3.20)
bagel(GtA)	2	17	25	36	24	20.80 (+3.40)
TokenFlow-XL(direct)	12	13	16	24	19	16.80 (-0.60)
show-o2(direct)	11	22	27	24	28	22.40 (+5.00)
show-o2(GtA)	6	18	30	15	23	18.40 (+1.00)
ILLUME+(direct)	4	20	28	29	14	19.00 (+1.60)
ILLUME+(GtA)	14	16	29	28	16	20.60 (+3.20)

Implications for RQ1. Our findings demonstrate that G2U is fundamentally a challenge of *representation coupling*. Simply injecting a generative pathway into a VLM does not automatically yield a stronger reasoner. Unification only benefits reasoning when the generative and understanding objectives are coherently aligned (as seen in spatial tasks); otherwise, the added generative capacity acts as an interfering auxiliary objective, diluting the model’s core cognitive reasoning power.

5.2 COMPARISON BETWEEN GTA AND DIRECT INFERENCE (RQ2)

Table 6: **Direct vs. GtA accuracy on generation-friendly subtasks.** GtA shows consistent improvements in multi-step spatial reasoning regimes. Results are formatted as (Direct / GtA).

Model	MSR	Maze	Sliding
	(Dir / GtA)	(Dir / GtA)	(Dir / GtA)
OmniGen2	0.220 / 0.290	0.043 / 0.046	0.000 / 0.001
OneCAT-3B	0.320 / 0.390	0.084 / 0.074	0.112 / 0.031
Ovis-U1	0.120 / 0.270	0.134 / 0.003	0.334 / 0.007
MIO	0.210 / 0.320	0.000 / 0.000	0.000 / 0.000
Bagel	0.260 / 0.250	0.021 / 0.281	0.009 / 0.195
Show-o2	0.270 / 0.300	0.142 / 0.178	0.137 / 0.183
ILLUME+	0.280 / 0.290	0.069 / 0.129	0.040 / 0.000

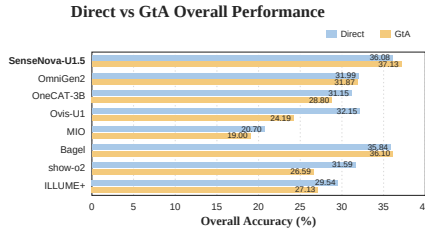


Figure 8: **GtA vs. Direct (Overall).** Overall accuracy comparison for the selected models supporting both inference protocols.

To investigate whether explicit intermediate visual generation provides additional benefits over direct unified inference, we evaluate each unified model under both the **Direct** and **GtA** protocols with matched inference budgets.

Overall Trend: The Risk of Visual Error Propagation.

Takeaway 3: The explicitly generation-augmented paradigm (GtA) degrades performance across most logic-intensive tasks, as forced visual externalization often propagates errors.

At the category level (Fig. 9), a majority of points lie below the $y = x$ line, indicating that GtA inference does not consistently outperform Direct inference across task regimes. This trend is further

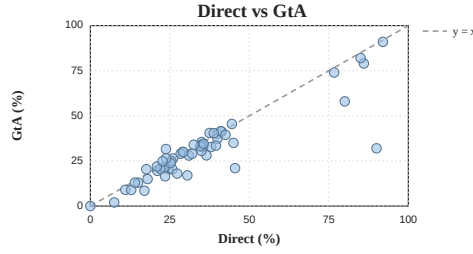


Figure 9: **GtA vs. Direct (Category-level)**. Each point represents a (model, category) pair comparing Direct and GtA accuracy.

Table 7: **Intermediate-image alignment across task categories**. Each cell reports the **RA / AL** scores (out of 5.0). Higher RA indicates stronger adherence to the intended visual reasoning scaffold; higher AL indicates tighter logical consistency between the final answer and the generated artifact.

Model	Real-world Apps	Geometry & Geometry	Physics	Puzzles & Games	Chart & Table	Spatial Intel.	Perception Reasoning	Overall (RA / AL)
Bagel	2.25/3.17	3.05/4.52	2.04/2.97	2.66/2.86	1.96/4.66	2.86/3.54	3.22/3.90	2.86/3.61
UAE	1.78/2.79	1.56/2.17	1.98/3.10	2.10/2.83	1.07/3.74	2.14/2.83	2.94/3.45	2.36/3.09
OmniGen2	1.51/2.90	1.26/1.56	1.49/2.31	1.88/2.40	1.34/2.97	1.79/2.70	2.13/3.22	1.86/2.79
GPT-4o + GPT-image	2.43/3.90	4.69/4.72	2.04/2.97	2.70/2.75	3.38/4.71	4.35/4.34	4.26/4.63	3.73/4.09
Ovis-U1	1.67/2.52	1.80/3.23	1.85/2.85	1.45/2.18	1.24/3.37	2.26/3.65	2.33/3.46	2.01/3.14
ILLUME+	1.39/2.95	1.41/2.71	1.34/2.08	2.01/2.50	1.01/4.41	1.42/2.05	2.59/3.30	2.00/2.84
OneCAT-3B	1.87/2.59	1.75/2.92	1.61/2.12	1.94/2.58	1.35/4.07	1.90/2.66	2.92/3.41	2.29/2.99
UniWorld-V1	1.62/2.88	1.48/2.15	1.74/2.76	1.60/2.41	1.38/3.58	2.01/3.12	2.36/3.44	1.98/3.04
MIO	1.92/2.70	2.42/2.88	1.98/2.91	2.24/2.36	1.95/3.82	2.65/3.22	3.18/3.76	2.67/3.24
Qwen2.5-7B + Qwen-Edit	1.68/2.95	1.32/1.88	1.55/2.48	1.72/2.46	1.47/3.44	1.92/3.08	2.41/3.55	1.99/3.05
Uni-Video	2.34/4.33	1.55/3.47	2.92/3.98	2.84/3.75	2.15/4.64	1.88/3.74	3.39/4.69	2.77/4.21
Show-o2	1.08/2.04	1.00/1.26	1.17/1.61	1.06/1.26	1.50/3.47	1.03/1.63	1.68/2.74	1.34/2.09
UniPic2	1.75/2.60	1.69/3.21	2.50/3.49	1.82/2.30	1.33/3.51	2.07/2.89	2.85/4.16	2.31/3.38
STAR-7B	2.11/3.60	1.54/2.44	2.37/3.50	1.85/2.43	2.77/3.72	2.22/3.53	2.85/3.75	2.39/3.36
X-Omni	1.78/2.71	1.11/1.29	2.21/2.95	1.97/2.37	1.75/2.45	1.60/2.18	1.85/2.42	1.80/2.35
SenseNova-U1.5-8B-MoT (GPT-5.5)	2.06/2.98	2.75/3.15	2.36/3.00	2.25/2.59	2.36/4.26	2.71/3.13	3.30/3.64	2.80/3.27

Evaluator note: SenseNova-U1.5-8B-MoT RA/AL scores were produced by GPT-5.5; all other models in this table were scored by GPT-4o. SenseNova means are computed over 2,995 valid examples; five policy-refused samples are excluded.

reflected in the aggregate comparison (Fig. 8), where GtA accuracy is frequently lower than Direct for models supporting both protocols. For several unified models (e.g., OneCAT-3B, Ovis-U1, and MIO), the degradation is substantial.

This overall decline highlights a critical vulnerability: explicitly externalizing intermediate visuals does not automatically improve reasoning. Instead, GtA prompting introduces severe failure modes when the generated artifact is inaccurate, semantically misaligned, or poorly integrated. From a coupling perspective, Direct mode allows the model to implicitly leverage generation-trained representations within its hidden space, completely avoiding the risk of committing to a flawed concrete image. In contrast, GtA forces a definitive visual hypothesis; if generative precision is lacking, this flawed visualization becomes a source of noise that misleads the subsequent answering module (see qualitative failure cases in Appendix D).

Transformation-Intensive Regimes: Visual CoT.

Takeaway 4: Explicit intermediate generation (GtA) yields noticeable performance gains in transformation-intensive scenarios where visual states serve as an external workspace.

Despite the overall drop, GtA is not uniformly detrimental. A fine-grained subtask analysis reveals that explicit generation yields consistent improvements in specific transformation-intensive regimes, such as **Maze navigation**, **Sliding puzzles**, and **Multi-Step Spatial Reasoning (MSR)** (Table 6).

These tasks share a defining structural property: they demand sequential, multi-step spatial manipulation and iterative state tracking. In such scenarios, explicit intermediate generation effectively functions as a *Visual Chain-of-Thought (CoT)*. By explicitly drawing out intermediate spatial configurations, the model decomposes complex reasoning trajectories into manageable sub-steps. The

generated artifact acts as an external cognitive workspace, stabilizing state tracking and significantly reducing the implicit memory burden required to track moving objects or paths across a long horizon.

Taken together, these findings indicate that the decisive factor for G2U success is not merely the act of explicit generation, but whether the generation functions as a reliable structural scaffold. Direct inference captures broad representational benefits, whereas GtA provides reasoning gains *only* when intermediate visuals are both structurally accurate and semantically aligned with sequential spatial logic.

5.3 CORRELATION ANALYSIS OF G2U GAINS ACROSS TASKS AND MODELS (RQ3)

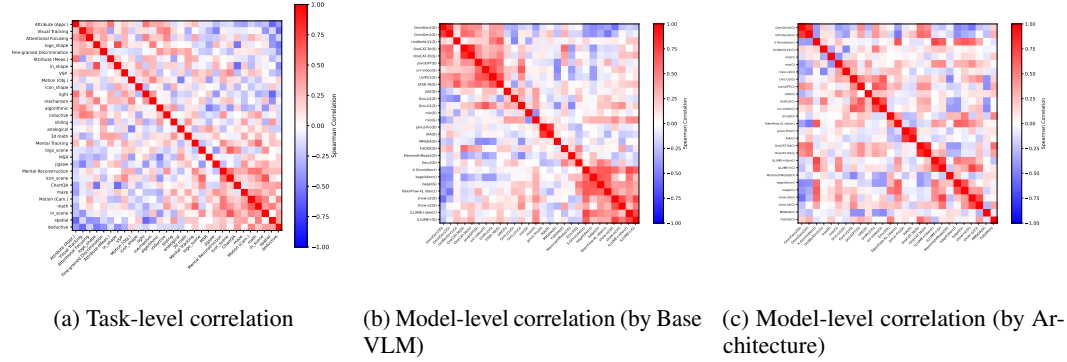


Figure 10: **Spearman correlation heatmaps of G2U gains (Δ)**. Rather than raw accuracy, these matrices visualize the correlation of the relative offset (Δ) to isolate generation-induced effects. (a) **Task-level**: Evaluates how different subtasks align in terms of G2U gains. The subtasks along the axes are roughly ordered from perception-oriented (left/top) to reasoning-oriented (right/bottom). Perception tasks and reasoning tasks each form distinctly positive internal clusters, yet they exhibit strong negative correlations with one another, highlighting a fundamental cognitive trade-off. (b) **Model-level (Base VLM)**: Models sharing the same foundational base weights exhibit highly similar G2U behavior, forming distinct high-correlation blocks. (c) **Model-level (Architecture)**: Grouping models purely by their generative paradigm (e.g., Diffusion vs. Autoregressive) yields weaker correlation patterns, indicating that inherited foundational representations govern G2U effects more strongly than the chosen generative architecture.

While RQ1 and RQ2 examine absolute performance and specific inference protocols, RQ3 investigates a deeper systemic question: *do the observed G2U gains (Δ) exhibit structured, predictable correlation patterns?*

Takeaway 5: At the task level, perception-oriented tasks and reasoning-oriented tasks each form internally correlated clusters, but typically exhibit negative correlations with one another. At the model level, unified models built upon the same base model demonstrate strong behavioral correlations, whereas those merely sharing architectural similarities exhibit much weaker correlations.

To rigorously answer this, we analyze the relative offset Δ (unified model minus its strictly paired base model). This isolation explicitly removes the confounding variable of underlying backbone strength, leaving only the pure generation-induced capability shift.

Task-Level Correlation We compute pairwise Spearman correlations between subtasks by treating each subtask as a vector of Δ values across all evaluated models. The resulting heatmap (Fig. 10a) reveals that G2U gains form distinct cognitive clusters. Subtasks leaning towards rigorous structural reasoning (e.g., jigsaw, maze) exhibit strong positive mutual correlations. Similarly, perception-oriented tasks (e.g., fine-grained recognition) also show a certain degree of internal similarity. Conversely, reasoning-oriented subtasks frequently exhibit *negative* correlations with perception-oriented tasks. This divergence provides compelling statistical evidence for the “alignment tax” discussed in RQ1: optimizing a model to benefit structural generation tasks often induces a representational trade-off that subtly degrades raw perceptual discrimination.

Model-Level Correlation We further compute correlations between models by treating each model as a vector of Δ across all subtasks. The model-level heatmaps (Fig. 10b and 10c) unearth a striking grouping effect. Unified models constructed upon the exact same foundational Base VLM (Fig. 10b) exhibit exceptionally strong correlations in their behavioral patterns, forming distinct high-correlation blocks. In stark contrast, models that merely share similar generative architectures (e.g., grouped by Autoregressive vs. Diffusion paradigms in Fig. 10c) demonstrate much weaker correlation signatures. This yields a critical insight for future unified model design: the mechanism by which generative capacity transfers to understanding is predominantly dictated by the *inherited foundational representations* (the VLM base), while merely adopting a similar generative architecture provides minimal behavioral consistency.

5.4 ARE INTERMEDIATE VISUALIZATIONS ALIGNED AND HELPFUL? (RQ4)

We diagnose the quality of explicitly generated intermediate images using the **RA** (Reasoning-to-Visual Alignment) and **AL** (Answer-to-Visual Alignment) metrics defined in Section 4.3. Detailed evaluation prompts and grading rubrics are provided in Appendix I. Here, we analyze the alignment patterns across task categories and models to mechanically explain the GtA effectiveness observed in RQ2.

Takeaway 6: High alignment fidelity is a necessary but not sufficient condition for G2U gains. While models draw faithfully in Perception tasks, explicit externalization is redundant; conversely, in structurally constrained domains, poor alignment leads to severe error propagation.

Two distinct patterns emerge from the diagnostic scores in Table 7, which perfectly corroborate our findings in RQ1 and RQ2.

First, **recognition-oriented tasks** (e.g., Perception Reasoning) universally achieve the highest RA/AL scores across almost all models. This indicates that current unified models can reliably generate visually faithful intermediate artifacts when the required transformation is a relatively direct visual translation. However, as established in RQ2, this high fidelity does not translate into performance gains. Because perception tasks are heavily reliant on low-level visual discrimination rather than multi-step structural logic, drawing the intermediate state is computationally redundant. The models “draw well” but do not actually need the drawings to answer.

Second, **logic-intensive and structurally constrained tasks** (such as Math & Geometry or specific physical simulations) exhibit significantly lower RA scores. This reflects the profound difficulty of faithfully externalizing complex, strict mathematical logic into a visual format. In these low-alignment settings, the generated auxiliary diagrams often omit critical constraints or distort geometric relationships. As a result, the subsequent answering module consumes a flawed visual premise (low AL), directly leading to the severe performance degradation (error propagation) observed in the GtA protocol.

Ultimately, the alignment diagnostic reveals that explicit generation is only beneficial when it operates in the “sweet spot”: tasks where visual externalization is structurally necessary to reduce cognitive load (like Spatial Intelligence, where RA scores are moderate but useful) *and* the model possesses sufficient alignment fidelity to produce trustworthy visual scaffolds.

6 CONCLUSION AND FUTURE WORK

In this work, we introduce **UniG2U**, a large-scale diagnostic benchmark for systematically studying when and how generation helps understanding in unified multimodal models. Across 3,000 curated instances spanning seven cognitive regimes, we show that generation-to-understanding (G2U) gains are highly task-dependent and architecture-dependent. Unified models do not consistently outperform their base VLMs; instead, improvements emerge primarily in spatial transformation and state-tracking tasks, where generation-trained representations regularize structural reasoning. Moreover, explicit Generate-then-Answer generation does not universally surpass direct unified inference. While intermediate visual externalization can assist reasoning, it often introduces new failure modes when generated artifacts are inaccurate or misaligned. Our correlation analyses further reveal that G2U gains follow structured task- and model-level patterns shaped by representational priors inherited from base models.

Looking forward, advancing G2U requires stronger integration mechanisms between generation and reasoning. Future directions include representation-level alignment objectives, reliability-aware or self-verifying intermediate generation, and closed-loop agentic refinement that mitigates error propagation. Extending beyond image-only intermediates to structured or multimodal reasoning artifacts may further enhance robustness. Additionally, investigating scaling laws of G2U effects and designing causal evaluation protocols to isolate representation changes remain important open problems. We hope UniG2U provides a principled foundation for future research toward unified multimodal systems in which generation and understanding mutually reinforce rather than interfere with one another.

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A PERFORMANCE ANALYSIS OF UNIFIED MODELS WITH FROZEN BACKBONES (NON-COT VERSION)

Table 8: **Direct (Non-CoT) results of frozen-backbone models on UniG2U.** All values are reported as percentages ($\times 100$). The Overall column represents the average accuracy across all task categories.

Model	Real-world Apps	Geometry Reasoning	Physics	Puzzles & Games	Chart	Spatial Intel.	Perception Reasoning	Overall
UniPic2(direct)	36.50	23.50	36.00	24.48	67.00	24.40	39.43	33.68
UniWorld-V1(direct)	35.50	22.00	33.00	25.62	65.00	27.40	36.66	32.79
STAR-7B(direct)	32.50	19.00	30.50	26.25	86.00	25.40	38.56	33.50
Uni-Video(direct)	30.00	20.50	29.50	27.16	81.00	26.00	39.67	33.93

Table 8 presents the Direct (Non-CoT) results of unified models that employ a frozen backbone. Theoretically, since the generative pathway is bypassed and the base vision-language model (VLM) backbone remains frozen during training, their performance should exactly match that of the base model (*qwen2.5-vl-7b*). However, empirical results show measurable deviations. We attribute these discrepancies to several implementation-level factors. First, subtle weight differences can emerge if a unified model utilizes Exponential Moving Average (EMA) checkpoints rather than the exact original base model weights, leading to slight performance drifts. Second, variations in inference hyperparameters—such as maximum generation length, input image resolution scaling, and decoding strategies—can significantly alter the final predictions. Third, inconsistencies in the software environment, such as different versions of PyTorch, CUDA, or the transformers library, often introduce deterministic execution variances. Furthermore, differences in visual pre-processing pipelines, particularly image encoding formats (e.g., reading raw images via PIL versus applying JPEG compression), affect pixel-level inputs. Finally, operating in different numerical precisions can cause compounding errors. For instance, computing Rotary Position Embeddings (RoPE) in bfloat16 instead of float32 can accumulate numerical errors over long sequences, eventually shifting the output logits and altering the results.

B ABLATION STUDY FOR MODELS WITHOUT EDIT TRAINING

Model	Real	Geometry	Physics	Puzzle	Chart	Spatial	Perception	Overall
UAE (Direct)	36.75	10.50	30.50	19.06	85.00	25.20	36.37	30.94
UAE (GtA)	23.00	7.50	32.50	20.33	71.00	22.80	34.68	28.61 (-2.33)
X-Omni (Direct)	21.50	9.50	29.50	26.24	83.00	27.00	35.31	30.86
X-Omni (GtA)	16.50	4.00	29.00	23.31	22.00	20.20	19.72	19.87 (-10.99)
TokenFlow-XL (Direct)	19.00	10.50	46.00	14.97	43.00	21.80	38.88	29.15
TokenFlow-XL (GtA)	14.50	9.50	26.50	10.06	38.00	21.80	34.28	24.50 (-4.65)

Table 9: Sheet7 results (values $\times 100$). Blue numbers indicate the overall drop relative to the corresponding Direct setting.

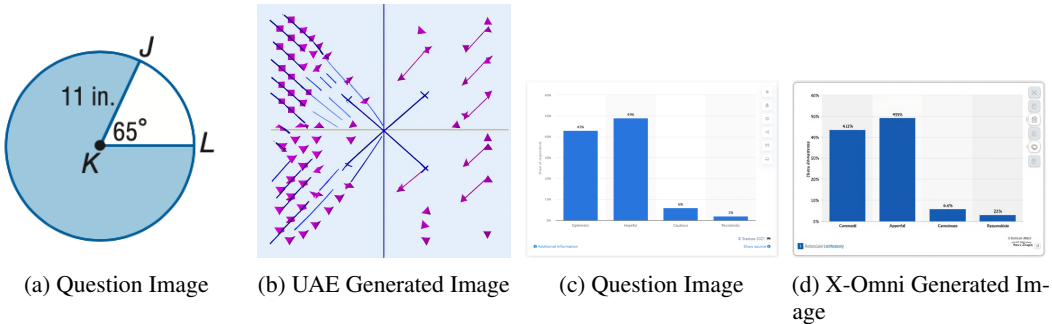


Figure 11: Examples of erroneous generation from edit-untrained models.

As discussed in Section 5, not all unified models evaluated in this work are trained with explicit image-to-image (edit) capabilities. For such models, Generate-then-Answer generation that requires intermediate image editing inherently falls outside their training distribution, effectively constituting an out-of-distribution (OOD) setting. Including their Generate-then-Answer results directly in the main comparison table could therefore introduce an unfair bias.

To better understand the effect of this OOD intermediate generation, we conduct additional ablation studies on a representative subset of edit-untrained models, with the results summarized in Table 9. Specifically, we categorize their Generate-then-Answer evaluation into two alternative variants:

(1) OOD-Image Generation (UAE and X-Omni). Despite lacking edit training, the first two models are forced to generate an intermediate image under the standard Generate-then-Answer protocol (i.e., conditioned on the original input image). The generated image is then re-consumed for answering. This setting measures how severely forcing OOD intermediate generation degrades downstream reasoning performance.

(2) Prompt-Only Intermediate Generation (TokenFlow-XL). Instead of feeding the original input image into the intermediate generation stage, we restrict the final model to producing an auxiliary image conditioned strictly on the textual generation prompt (i.e., no image-to-image conditioning). The generated image is subsequently used as additional context for answering. This setting isolates the effect of prompt-level generation without invoking explicit edit-style image transformations.

Findings. As demonstrated in Table 9, both Generate-then-Answer variants consistently lead to overall performance degradation compared to their Direct inference counterparts. Specifically, forcing OOD-Image generation causes overall performance drops of 2.33% and 10.99% for UAE and X-Omni, respectively, while relying on Prompt-Only generation leads to a 4.65% drop for TokenFlow-XL.

Furthermore, the qualitative examples presented in Figure 11 corroborate these quantitative declines. Visual inspection reveals that the intermediate images produced by these edit-untrained models are often completely chaotic, structurally irrelevant, or fail to encode the necessary information. Instead of aiding the reasoning process, these misaligned generations actively mislead the subsequent answering stage.

Overall, these ablation results validate our decision to exclude the Generate-then-Answer scores of edit-untrained models from the main comparison table. They strongly emphasize that Generate-to-Understand (G2U) gains rely critically on explicit and training-aligned intermediate generation capabilities.

C ADDITIONAL RESULT ANALYSIS: STRUCTURED SUBTASK GAINS

In this section, we analyze the complete experimental results (Table 11) to identify subtasks where generation-to-understanding (G2U) gains are particularly evident.

While overall improvements are not universal, we observe that unified models demonstrate clearer and more repeatable gains on specific structured subtasks. In particular, **Spatial Intelligence** and **illusion-related perceptual reasoning** subtasks show more consistent positive offsets compared to base VLMs. These regimes require transformation-aware reasoning, geometric consistency, or resolving visually ambiguous patterns where generation-trained representations appear to regularize internal spatial structure.

For example, in the *spatial* subtask under Perception Reasoning, multiple unified variants (e.g., UniWorld-V1, JarvisGPT, uni-video, STAR-7B) achieve higher scores than the corresponding base model. Similarly, illusion-sensitive discrimination subtasks (e.g., logo-related and shape-sensitive categories) exhibit non-trivial improvements in certain unified configurations. These gains suggest that generation-induced representation coupling may enhance the model’s robustness to visually ambiguous or structurally complex inputs.

In contrast, purely recognition-heavy subtasks (e.g., *icon_scene*, *in_scene*) do not show systematic improvement and sometimes experience mild degradation. This further supports our central claim: G2U benefits are not uniform but emerge in transformation-centric or structurally constrained regimes.

Table 11 provides the detailed subtask-level results supporting this analysis.

D FAILURE CASE TAXONOMY FOR GENERATION-AUGMENTED UNDERSTANDING

In this appendix, we present representative failure cases of generation-augmented reasoning. We organize all examples under a unified three-way taxonomy based on the relationship between generated diagrams and downstream reasoning performance.

Table 10: **Absolute results on selected Illusion and Spatial subtasks.** All values are reported as percentages ($\times 100$). Baseline models (marked with *) are highlighted with gray background. **Overall** is computed as the average over the 5 shown subtasks, and the value in parentheses denotes the gap to the corresponding baseline within each model block. The best **Overall** within each block is **bolded**.

Model	Illusion		Spatial Intelligence			Overall (Δ)
	icon_shape	in_shape	MSR	Attr. (Meas.)	Motion (Cam.)	
qwen2.5-vl-7b*	5	10	25	20	29	17.80 (+0.00)
OmniGen2(direct)	6	8	22	33	12	16.20 (-1.60)
OmniGen2(GtA)	6	11	29	33	18	19.40 (+1.60)
UniWorld-V1(GtA)	4	10	27	22	21	16.80 (-1.00)
OneCAT-3b(direct)	2	8	32	30	24	19.20 (+1.40)
OneCAT-3b(GtA)	1	7	39	28	3	15.60 (-2.20)
JavisGPT(direct)	12	13	16	24	19	16.80 (-1.00)
uni-video(GtA)	18	36	23	18	10	21.00 (+3.20)
UniPic2(GtA)	8	24	21	23	23	19.80 (+2.00)
STAR-7B(GtA)	7	13	27	22	21	18.00 (+0.20)
Qwen2.5-VL-3B*	6	9	22	32	18	17.40 (+0.00)
UAE(direct)	7	8	26	22	20	16.60 (-0.80)
Ovis-U1(direct)	7	10	12	28	18	15.00 (-2.40)
Ovis-U1(GtA)	5	15	27	27	11	17.00 (-0.40)
yi-6B-vl*	17	16	24	26	11	18.80 (+0.00)
mio(direct)	5	13	21	24	23	17.20 (-1.60)
mio(GtA)	15	16	32	24	22	21.80 (+3.00)
deepseek-v2*	8	13	23	19	14	15.40 (+0.00)
Janus-Pro(direct)	4	21	28	37	16	21.20 (+5.80)
AIA(direct)	5	18	23	43	19	21.60 (+6.20)
LLaDA-Instruct*	0	8	21	41	11	16.20 (+0.00)
MMaDA(direct)	1	3	32	27	20	16.60 (+0.40)
Janus-1.3B*	6	10	15	28	18	15.40 (+0.00)
FUDOK(direct)	10	5	29	28	29	20.20 (+4.80)
Qwen3-VL-8B*	13	22	23	35	20	22.60 (+0.00)
MammothModa2(direct)	5	10	14	19	18	13.20 (-9.40)
llava-hf*	10	20	28	24	14	19.20 (+0.00)
Emu3(direct)	0	2	31	23	14	14.00 (-5.20)
llava_onevision*	6	18	17	26	20	17.40 (+0.00)
X-Omni(direct)	5	11	20	31	24	18.20 (+0.80)
bagel(direct)	2	18	26	31	26	20.60 (+3.20)
bagel(GtA)	2	17	25	36	24	20.80 (+3.40)
TokenFlow-XL(direct)	12	13	16	24	19	16.80 (-0.60)
show-o2(direct)	11	22	27	24	28	22.40 (+5.00)
show-o2(GtA)	6	18	30	15	23	18.40 (+1.00)
ILLUME+(direct)	4	20	28	29	14	19.00 (+1.60)
ILLUME+(GtA)	14	16	29	28	16	20.60 (+3.20)

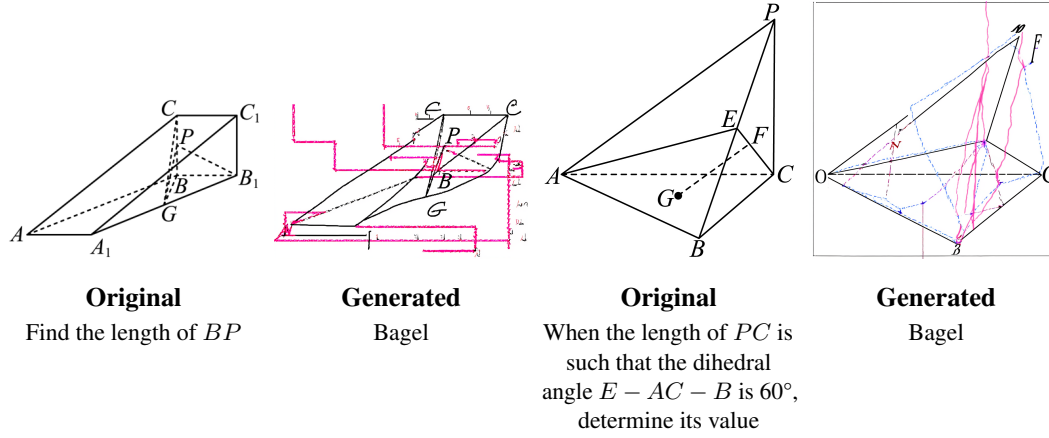


Figure 12: Capability failure. The model produces geometrically inconsistent or physically invalid diagrams, preventing correct downstream reasoning.

Why failure cases matter. Generation-augmented understanding is often motivated by the intuition that “drawing helps thinking”: a faithful diagram can externalize latent structure, reduce working-memory burden, and provide geometric/physical constraints that guide computation. However, *visual generation is itself a structured prediction problem*. When the generated artifact is even slightly inconsistent with the underlying task, the downstream solver may inherit or amplify those errors (e.g., by trusting a distorted geometry, an incorrect scale, or a spurious path). Thus, it is important to characterize *how* generation fails, not merely *whether* an answer is wrong.

Selection protocol. We curate cases from diverse domains (geometry, physics, charts, and factual QA) to ensure the taxonomy covers (i) tasks where diagrams are essential (geometry/physics), (ii) tasks where diagrams can be helpful but must encode precise structure (charts), and (iii) tasks where diagrams are unnecessary (pure textual knowledge). For each case, we pair the original prompt with a model-generated visual output and highlight the key mismatch that explains downstream failure.

Three criteria: validity, relevance, and utility. Across domains, successful generation must satisfy three requirements: (i) **structural validity** (the diagram is internally consistent and respects physical/geometric rules), (ii) **semantic relevance** (the content corresponds to the task entities and relations), and (iii) **reasoning utility** (the diagram adds constraints or reduces ambiguity in a way that supports solving). Our taxonomy maps naturally to violations of these criteria: Category I breaks validity, Category II satisfies relevance but lacks utility, and Category III fails relevance altogether.

Category I: Capability Failure. The model fails to generate structurally correct or quantitatively valid diagrams. Rendered visuals are geometrically distorted, structurally inconsistent, or physically invalid, making downstream reasoning unreliable or impossible. This category is the most direct failure mode: even if the image appears “diagram-like”, it violates invariants required by the task.

Common symptoms.

- *Geometric inconsistency*: angles/parallel lines/circle tangency implied by the prompt are not preserved.
- *Quantitative distortion*: lengths, slopes, or axis scales are visually inconsistent with stated values.
- *Physical invalidity*: forces/velocities are drawn with wrong directions or incompatible magnitudes.
- *Missing constraints*: key elements (labels, right-angle marks, reference points) are omitted or misplaced.

Why it breaks reasoning. Even a strong textual solver may implicitly treat the diagram as ground truth. When the generated visual is invalid, the solver can be pushed toward incorrect intermediate computations (e.g., wrong triangle similarity, wrong area under a curve, wrong cross-product direction), or may be forced to ignore the diagram entirely, losing the intended benefit of externalization.

Figure 12 shows two geometry instances where the generated diagrams violate the required geometric/physical constraints.

Figure 13 presents two physics cases where the generated visuals fail to preserve quantitative/structural relations needed for integration- or geometry-based computation.

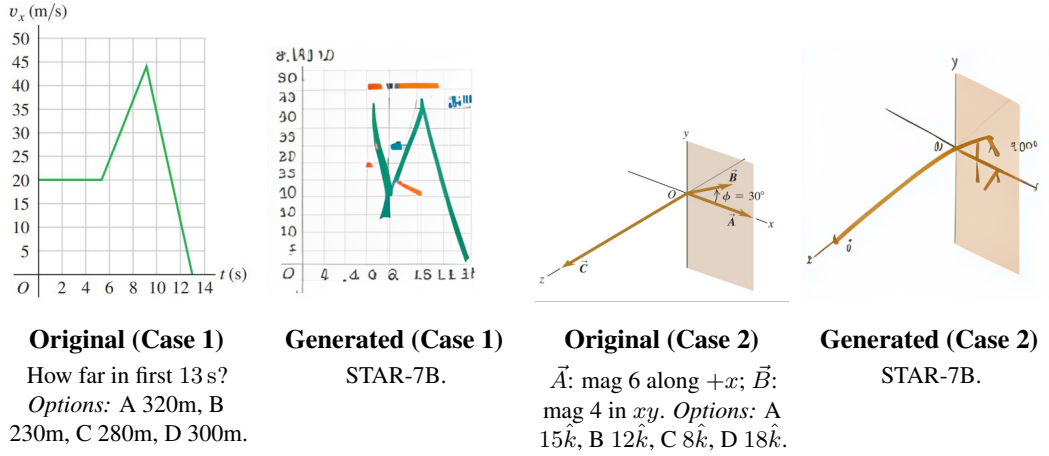


Figure 13: Capability failure in physics reasoning. Generated diagrams fail to preserve quantitative/structural relations required for correct integration- or geometry-based computation.

Implication. Category I indicates a bottleneck in the generator’s *diagram grammar*: models may learn surface styles (lines, arrows, labels) without learning domain invariants. A practical mitigation is to introduce lightweight validation or self-checking (e.g., verify right angles, axis monotonicity, connectivity constraints) before passing the diagram to the solver, or to route such instances back to a pure-text solution when validity is uncertain.

Category II: Surface-Relevant but Non-Utility Generation. The model generates diagrams that are semantically related to the prompt, yet they add no constraints or computational benefit for solving. In these cases, the generation is “on-topic” but *not instrumental*: the diagram either restates what is already explicit in text, or fails to encode the specific structure needed.

Common symptoms.

- *Correct but redundant*: the diagram matches the description but does not reduce uncertainty.
- *Missing operational structure*: key relations for computation (e.g., graph scales, maze connectivity) are absent.
- *Aesthetic completion*: the model fills the page with plausible visuals rather than task-sufficient constraints.

Why it fails to help. Utility is not guaranteed by relevance: even a perfectly relevant illustration can be useless if it does not support the specific operation needed (e.g., reading off a slope, tracing a path, or identifying a computable sub-structure). This failure mode is particularly important because it can *waste* generation budget without improving accuracy, and may even distract downstream reasoning by introducing extra, non-actionable details.

Figure 14 shows cases where the output looks on-topic but does not help: one is correct yet uninformative, and the other lacks valid connectivity/path structure.

Figure 15 illustrates surface-triggered generation in kinematics: lexical cues (e.g., “coin”) induce drawing, but the task requires computation, so the visuals contribute little to reasoning.

Implication. Category II suggests that “draw something relevant” is an insufficient objective. For G2U, the generator must be *goal-conditioned*: it should produce the minimal structure that makes the answering computation easier (e.g., annotated axes, explicit path graph, marked equal angles), rather than a generic illustration. This also motivates *budget-aware routing*: when predicted utility is low, skipping generation can improve value-per-token without sacrificing accuracy.

Category III: Irrelevant Generation. The generated image is not meaningfully related to the task, indicating complete generation–task misalignment. Here, generation is triggered when it should not be (e.g., factual QA), or the model produces a visual that does not encode the required structured content (e.g., ChartQA).

Common symptoms.

- *Mode confusion*: the model treats a purely textual query as an invitation to draw.

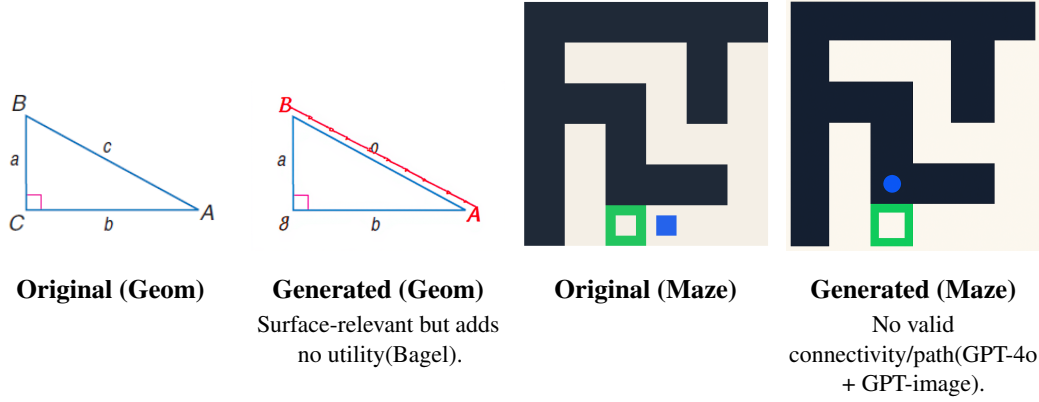


Figure 14: Surface-relevant but non-utility generation. Outputs are related to the prompt but provide no additional constraints or computational benefit.

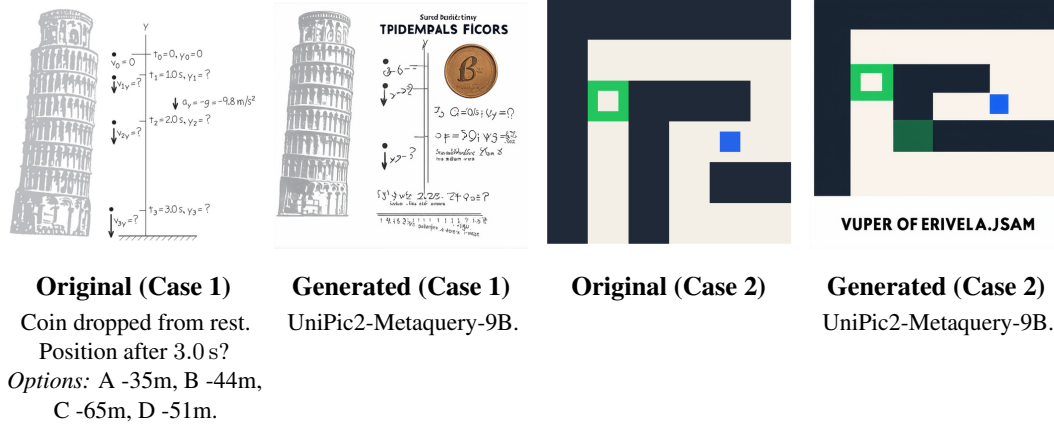


Figure 15: Surface-triggered generation. The model draws a “coin” due to lexical cues, while the task requires kinematic computation; the generated visuals provide little to no reasoning utility.

- *Structure loss*: the output resembles an image but lacks the schema needed (bars/axes/labels) to be a chart.
- *Hallucinated content*: visuals introduce unrelated entities, styles, or scenes not grounded in the prompt.

Why it hurts. Irrelevant generation is harmful beyond wasted compute: it can shift the solver’s attention to spurious cues, reduce trust in the system, and complicate debugging (the output looks “confident” but is unrelated). In knowledge QA, the correct behavior is often to answer directly; in chart reasoning, the correct behavior is to preserve structured data rather than produce decorative images.

Figure 16 shows irrelevant outputs in chart and factual QA settings: chart generations fail to encode structured data, while factual questions unnecessarily trigger visual generation.

Figure 17 provides additional knowledge-QA examples where the model produces visual content despite the task being purely factual.

Practical lesson for G2U system design.

1. **Validate** when diagrams are intended to be constraints (Category I risk).
2. **Predict utility** and skip redundant illustrations (Category II risk).
3. **Route by modality** and avoid image generation for pure text tasks (Category III risk).

E UNI-MMMU TASK DESIGN

Uni-MMMU consists of three task types: Jigsaw, Maze, and Sliding Puzzle. Each task evaluates different aspects of visual reasoning through interleaved image-text generation. (following case from Model - Nano Banana)

E.1.2 VISUAL COT GENERATION FLOW

Step 1 - Input Images:



Figure 18: Jigsaw task input images

Step 2 - Generate Completion with Candidate 0:

Model generates a 2x2 image with Candidate 0 placed in the bottom-right cell. The other three cells remain pixel-identical to the reference.



Figure 19: Generated image: Completion with Candidate 0

Step 3 - Generate Completion with Candidate 1:

Model generates a 2x2 image with Candidate 1 placed in the bottom-right cell. The other three cells remain pixel-identical to the reference.

Step 4 - Generate Final Answer:

Model analyzes both generated completions and outputs:

```
<FINAL_ANSWER_JSON>
{"choice": 0, "rationale": "Candidate 0 shows better seam continuity"}
</FINAL_ANSWER_JSON>
```

E.2 MAZE SOLVING TASK

E.2.1 TASK DESCRIPTION

Input: Maze image with:

- Black squares: walls (impassable)



Figure 20: Generated image: Completion with Candidate 1

- White squares: paths (walkable)
- Blue dot: starting position
- Green frame: goal region

Task: Find a sequence of moves from start to goal.

Constraints: Legal moves: up, down, left, right; one cell per step; no diagonals; no wall crossing.

Output: <ANSWER_JSON>["right", "down", ...]

Metrics: maze_text_exact, maze_text_frame_acc

Dataset: 100 samples

E.2.2 VISUAL CoT GENERATION FLOW

Step 1 - Input Image:

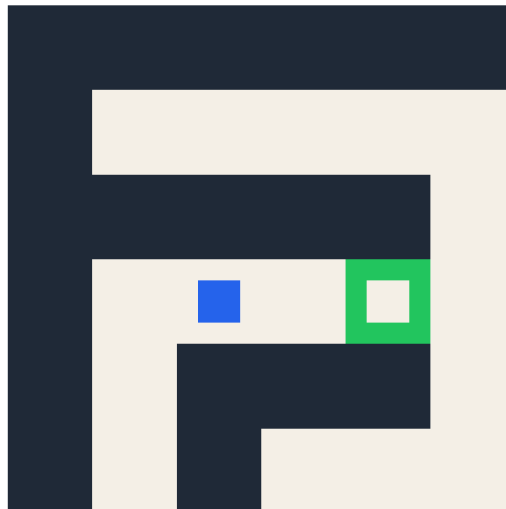


Figure 21: Initial maze state (blue dot at start, green frame marks goal)

Step 2 - Generate Step-by-Step Images:

For each move in the solution, the model generates an image showing the maze state after that move. Only the blue dot position changes; all other elements remain identical.

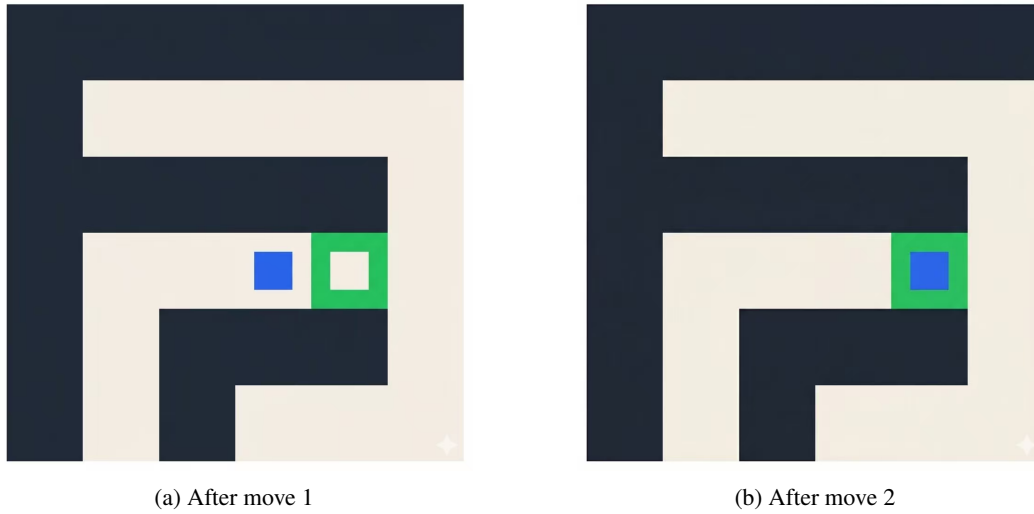


Figure 22: Example step images (blue dot moves through the maze)

Step 3 - Generate Final Answer:

After generating all step images, the model outputs:

```
<ANSWER_JSON>["right", "down", "right", ...]</ANSWER_JSON>
```

Key Constraints:

- Number of generated images must equal number of moves
- Each image shows state after exactly one move
- Do not include initial state
- No collages, arrows, or annotations

E.3 SLIDING PUZZLE TASK**E.3.1 TASK DESCRIPTION****Input:**

- Initial state: 3x3 grid with 8 colored tiles and 1 empty space (red)
- Goal state: Solved puzzle configuration

Task: Find a sequence of moves to solve the puzzle.

Move Semantics: Moves are named by the direction the colored tile moves (not the empty space). Example: tile above empty moving down = "down".

Constraints: Legal moves: up, down, left, right; one tile per step.

Output: <ANSWER_JSON>["down", "right", ...]

Metrics: sliding_text_exact, sliding_text_frame_acc

Dataset: 54 samples

E.3.2 VISUAL CoT GENERATION FLOW**Step 1 - Input Images:****Step 2 - Generate Step-by-Step Images:**

For each move in the solution, the model generates an image showing the puzzle state after that move. Only tile positions change; colors remain identical.



Figure 23: Sliding puzzle input: initial and goal states

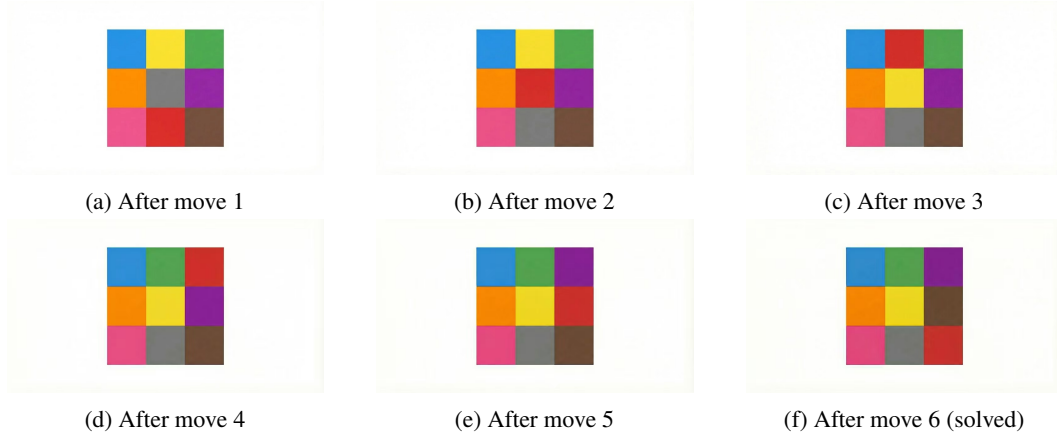


Figure 24: Example step-by-step puzzle solving sequence

Step 3 - Generate Final Answer:

After generating all step images, the model outputs:

```
<ANSWER_JSON>["down", "right", "up", "left", "down", "right"]</ANSWER_JSON>
```

Key Constraints:

- Number of generated images must equal number of moves
- Each image shows state after exactly one move
- Do not include initial state
- No collages or multi-panel images
- Tile colors must remain pixel-identical

F EXPERIMENTAL RESULTS

G TASK PROMPTS (NON-COT VERSION)

G.1 MMSI-BENCH TASKS

G.1.1 MMSI_ATTRIBUTE_APPR (ATTRIBUTE - APPEARANCE) AND MMSI_ATTRIBUTE_MEAS (ATTRIBUTE - MEASURABLE) AND MMSI_MSR (MULTI-STEP REASONING)

```
pre_prompt: "Pay close attention to the visual appearance attributes
(color, shape, texture, orientation, count, visual features) in the
images. Observe details carefully to identify and count objects.\n\n"
```

```
post_prompt: "\n\nBased on your visual observation, answer with the
```

Table 11: Main results on UniG2U across categories (Part 1/2). All values are reported as percentages ($\times 100$). Baseline models (marked with *) are highlighted with gray background.

Model	Perception Reasoning												Real-world Apps			Chart
	icon_scene	icon_shape	in_scene	in_shape	logo_scene	logo_shape	algorithmic	analogical	deductive	inductive	spatial	Fine Discrim.	Attn. Focus.	VSP Reason.		ChartQA
qwen2.5-vl-7b*	98	5	98	10	94	20	31	24	46	26	28	11	44	36.5	86	
OmniGen2(direct)	89	6	88	8	89	18	33	20	39	18	24	11	50	32.5	86	
OmniGen2(GA)	80	6	81	11	78	23	28	27	39	25	22	9.82	49	33.5	79	
UniWorld-V1(GA)	99	4	99	10	95	19	36	22	47	21	34	6.75	14	34	80	
OneCAT-3b(direct)	85	2	84	8	89	13	30	29	38	20	22	10.02	46	33	76.68	
OneCAT-3b(GA)	87	1	82	7	90	4	23	29	35	21	24	9.2	33	34	74	
JavisGPT(direct)	86	12	75	13	78	15	36	23	43	22	32	8.32	36	28	78.6	
uni-video(GA)	97	18	96	36	79	50	32	21	37	26	32	9.82	39	29	61	
UniPic2(GA)	94	8	88	24	90	22	32	22	40	23	28	11.66	21	14	47	
STAR-7B(GA)	95	7	98	13	93	19	32	21	46	23	29	6.75	35	33	79	
Qwen2.5-VL-3B*	89	6	87	9	90	18	30	22	44	22	21	6.75	51	31	87	
UAE(direct)	89	7	88	8	90	18	31	32	34	26	20	10	45	28.5	85	
Ovis-U1(direct)	91	7	83	10	87	18	28	27	30	21	27	7.36	59	32	90	
Ovis-U1(GA)	75	5	73	15	48	21	30	22	30	27	30	6.13	23	19	32	
yi-6B-vl*	81	17	20	16	84	23	31	18	27	25	33	11.66	27	21	19	
mio(direct)	46	5	36	13	30	27	32	22	35	28	24	14.72	35	1	15	
mio(GA)	53	15	36	16	28	19	21	23	32	28	27	10.43	30	0	13	
deepseek-v2*	60	8	78	13	90	19	40	23	35	22	34	12.88	22	0	74	
Janus-Pro(direct)	82	4	85	21	74	32	26	21	30	25	27	9.82	27	28	29	
Ala(direct)	85	5	87	18	81	28	31	22	24	31	29	11.04	25	2	33	
LLaDA-Instruct*	7	0	9	8	7	5	29	24	32	24	24	3.07	32	22.5	9	
MMaDA(direct)	55	1	47	3	44	4	25	29	31	18	22	5	32	25.5	14	
Janus-1.3B*	57	6	35	10	35	23	21	23	25	22	22	11.04	26	1	50	
FUDOK(direct)	10	10	10	5	11	4	25	24	32	28	24	8	34	1	22	
Qwen3-VL-8B*	100	13	100	22	97	28	45	29	40	21	32	15	48	32	89	
MammothModu2(direct)	98	5	98	10	94	20	37	26	48	20	30	4.91	45	22	91	
llava-hf*	84	10	81	20	59	31	23	30	29	20	22	9.2	27	19	28	
Emu3(direct)	51	0	60	2	35	5	33	25	34	24	28	9.82	31	2	67	
llava_onevision*	86	6	74	18	91	34	35	25	28	32	22	11.04	54	32	78	
X-Omni(direct)	91	5	97	11	85	13	27	24	35	17	27	8.59	42	24	83	
bagel(direct)	91	2	88	18	91	20	31	26	46	42	35	9	50	26	92	
bagel(GA)	81	2	81	17	80	20	30	26	43	40	35	9.82	46	19.5	91	
TokenFlow-XL(direct)	86	12	75	13	78	15	36	23	43	22	32	8.32	36	28	78.5	
show-o2(direct)	89	11	80	22	81	37	22	24	36	21	26	7.36	38	33	45	
show-o2(GA)	56	6	51	18	53	31	21	23	32	23	25	9.82	35	31	35	
ILLUME+(direct)	86	4	84	20	91	26	22	28	30	18	23	6.75	36	13	80	
ILLUME+(GA)	78	14	83	16	87	21	18	27	37	17	25	4.29	21	20	58	
gpt4o	96	16	93	28	100	33	37	32	42	27	32	10	53	37.5	58	
ovis2.5	99	8	92	21	97	27	34	34	49	25	32	9.82	57	31	89	
GPT-4o + GPT-image	90	17	89	25	92	30	35	27	32	30	19	15.95	47	18.5	60	
gemini pro	100	38	100	43	100	58	79	73	50	31	62	23.31	64	79	68	
qwenedit	99	5	98	11	89	25	24	21	45	27	31	7.36	42	25.5	80	
SenseNova-U1.5-8B-MoT (Direct)	96	10	96	15	93	12	30	21	39	27	35	10	57	28	85	
SenseNova-U1.5-8B-MoT (Visual CoT)	95	8	91	26	90	21	33	26	35	27	34	15	51	28	82	

Table 12: Main results on UniG2U across categories (Part 2/2). Continuing from Part 1 with the remaining columns.

Model	Geometry Reasoning			Spatial Intel.				Puzzles & Games				Physics		
	Geometry	3d Geometry	MSR	Attr. (Means)	Attr. (Appr.)	Motion (Cam.)	Motion (Obj.)	Mental Recon.	Mental Track.	Visual Track.	mare	jigsaw	sliding	light mechanism
qwen2.5-vl-7b*	46	0	25	20	10	29	33	36	25	9.64	14.12	52	5.61	34
OmniGen2(direct)	28	0	22	33	19	12	33	31	23	7.23	4.31	48	0	42
OmniGen2(GA)	26	0	29	33	17	18	35	31	23	6.02	4.6	54	0.13	41
UniWorld-V1(GA)	38	0	27	22	16	21	38	39	24	6.02	14.43	47	6.14	33
OneCAT-3b(direct)	34	0	32	30	27	24	13	24	25	6.02	8.4	50	11.18	41
OneCAT-3b(GA)	17	0	39	28	23	3	26	23	24	6.02	7.43	50	3.1	37
JavisGPT(direct)	42	0	16	24	18	19	31	30	33	8.74	0.48	51	10.44	25
uni-video(GA)	43	0	23	18	17	10	32	28	26	9.64	21.9	57	30.79	31
UniPic2(GA)	33	1	21	23	21	23	35	33	22	4.82	11.53	48	18.71	28
STAR-7B(GA)	29	0	27	22	17	21	39	31	27	6.02	6.87	46	19.72	34
Qwen2.5-VL-3B*	31	0	22	32	24	18	36	31	24	3.61	1.64	50	0	42
UAE(direct)	21	0	26	22	26	20	32	33	19	4.82	5.17	38	5.91	35
Ovis-U1(direct)	47	0	12	28	7	18	23	31	28	8.43	13.44	49	33.39	35
Ovis-U1(GA)	33	0	27	27	15	11	22	22	19	3.61	0.25	52	0.74	19
yi-6B-vl*	0	0	24	26	22	11	30	33	22	6.02	0	11	0	35
mio(direct)	0	0	21	24	24	23	25	25	20	3.61	0	21	0	30
mio(GA)	0	0	32	24	14	22	16	27	17	3.61	0	1	0	27
deepseek-v2*	9	0	23	19	9	14	22	11	1	6.02	11.2	53	28.82	21
Janus-Pro(direct)	15	0	28	37	25	16	15	21	26	3.61	12.53	54	7.87	21
Ala(direct)	17	0	23	43	21	19	27	31	19	4.82	12.87	51	19.93	31
LLaDA-Instruct*	3	0	21	41	18	11	26	22	26	3.61	12.87	48	0	24
MMaDA(direct)	10	0	32	27	25	20	21	21	28	2.41	0	35	0	28
Janus-1.3B*	5	0	15	28	21	18	27	24	19	2.41	1.81	18	0.26	19
FUDOK(direct)	2	0	29	28	36	29	23	22	25	3.61	0	0	0	25
Qwen3-VL-8B*	67	11	23	35	22	20	30	32	26	8.43	15.53	45	8.59	37
MammothModu2(direct)	0	0	14	19	18	18	40	35	24	7.23	0	0	0	40
llava-hf*	0	0	28	24	30	14	27	26	20	4.82	0	50	3.35	21
Emu3(direct)	2	0	31	23	28	14	20	33	27	4.82	1.39	0	11.85	25
llava_onevision*	21	0	17	26	29	20	35	31	23	16.87	7.46	51	9.25	42
X-Omni(direct)	19	0	20	31	26	24	34	30	27	6.02	12.87	60	11.23	35
bagel(direct)	58	0	38	30	20	26	39	37	30	7.23	2.15	52	0.9	35
bagel(GA)	58	2	25	36	27	24	33	41	31	4.82	28.1	55	19.45	38
TokenFlow-XL(direct)	42	0	16	24	18	19	31	30	33	8.74	0.48	51	10.44	25
show-o2(direct)	22	0	27	24	21	28	29	32	22	9.64	14.24	56	13.74	37
show-o2(GA)	18	0	30	15	0	23	34	30	23	10.84	17.84	52	18.32	43
ILLUME+(direct)	15	0	28	29	14	14	31	27	19	6.02	6.93	53	4.02	38
ILLUME+(GA)	4	0	29	28	15	16	17	20	18	2.41	12.87	52	0	37
gpt4o	53	1	33	38	32	34	43	33	24	14.46	32.15	52	21.99	36
ovis2.5	64	1	27	40	21	22	29	25	23	8.43	15.07	45	16.93	42
GPT-4o + GPT-image	47	7	23	30	23	23	36	36	20	16.87	39.84	50	29.4	37
gemini pro	86	84	35	60	29	28	40	39	70	26.51	49.29	77	73.57	89
qwenedit	34	0	19	19	15	18	37	32	23	7.23	25.57	52	19.7	33
SenseNova-U1.5-8B-MoT (Direct)	55	34	32	34	18	27	35	26	25	10.84	0	62	0	27
SenseNova-U1.5-8B-MoT (Visual CoT)	65	26	30	37	18	23	42	29	28	12.05	4.33	62	0	36

Table 13: **Delta results on UniG2U across categories (Sheet 3, Part 1/2).** Baseline models (marked with *) are highlighted with gray background. A dash denotes either a missing value or an unavailable delta; SenseNova-U1.5 is unpaired and therefore has no delta.

Model	Perception Reasoning												Real-world Apps			Chart
	icon_scene	icon_shape	in_scene	in_shape	logo_scene	logo_shape	algorithmic	analogical	deductive	inductive	spatial	Fine Discrim.	Attn. Focus.	VSP Reason.	ChartQA	ChartQA
qwen2.5-vl-7b*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OmniGen2(direct)	-9	1	-10	-2	-5	-2	2	-4	-7	-8	-4	0	6	-4	0	-
OmniGen2(GtA)	-18	1	-17	1	-16	3	-3	3	-7	-1	-6	-1.18	5	-3	-7	-
UniWorld-V1(GtA)	1	-1	1	0	1	-1	5	-2	1	-5	6	-4.25	-30	-2.5	-6	-
OncCAT-3b(direct)	-13	-3	-14	-2	-5	-7	-1	5	-8	-6	-6	-0.98	2	-3.5	-9.32	-
OncCAT-3b(GtA)	-11	-4	-16	-3	-4	-16	-8	5	-11	-5	-4	-1.8	-11	-2.5	-12	-
JavisGPT(direct)	-12	7	-23	3	-16	-5	5	-1	-3	-4	4	-2.68	-8	-8.5	-7.4	-
uni-video(GtA)	-1	13	-2	26	-15	30	1	-3	-9	0	4	-1.18	-5	-7.5	-25	-
UniPic2(GtA)	-4	3	-10	14	-4	2	1	-2	-6	-3	0	0.66	-23	-22.5	-39	-
STAR-7B(GtA)	-3	2	0	3	-1	-1	1	-3	0	-3	1	-4.25	-9	-3.5	-7	-
Qwen2.5-VL-3B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
UAE(direct)	0	1	1	-1	0	0	1	10	-10	4	-1	3.25	-6	-2.5	-2	-
Ovis-U1(direct)	2	1	-4	1	-3	0	-2	5	-14	-1	6	0.61	8	1	3	-
Ovis-U1(GtA)	-14	-1	-14	6	-42	3	0	0	-14	5	9	-0.62	-28	-12	-55	-
yi-6B-vl*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
miu(direct)	-35	-12	16	-3	-54	4	1	4	8	3	-9	3.06	8	-20	-4	-
miu(GtA)	7	10	0	3	-2	-8	-11	1	-3	0	3	-4.29	-5	-1	-2	-
deepseek-v2*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Janus-Pro(direct)	22	-4	7	8	-16	13	-14	-2	-5	3	-7	-3.06	5	28	-45	-
ALA(direct)	25	-3	9	5	-9	9	-9	-1	-11	9	-5	-1.84	3	2	-41	-
LLaDA-Instruct*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MMaDA(direct)	48	1	38	-5	37	-1	-4	5	-1	-6	-2	1.93	0	3	5	-
Janus-1.3B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
FUDOK(direct)	-47	4	-25	-5	-24	-19	4	1	7	6	2	-3.04	8	0	-28	-
Qwen3-VL-8B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MammothMod2(direct)	-2	-8	-2	-12	-3	-8	-8	-3	8	-1	-2	-10.09	-3	-10	2	-
llava-h*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Emu3(direct)	-33	-10	-21	-18	-24	-26	10	-5	5	4	6	0.62	4	-17	39	-
llava_onevision*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
X-Omni(direct)	5	-1	23	-7	-6	-21	-8	-1	7	-15	5	-2.45	-12	-8	5	-
bagel(direct)	5	-4	14	0	0	-14	-4	1	18	10	13	-2.04	-4	-6	14	-
bagel(GtA)	-5	-4	7	10	17	-5	-1	15	8	13	-5	-1.22	-8	-12.5	13	-
TokenFlow-XL(direct)	0	6	1	-5	-13	-19	1	-2	15	-10	10	-2.72	-18	-4	0.6	-
show-o2(direct)	3	5	6	4	-10	3	-13	-1	8	-11	4	-3.68	-16	1	-33	-
show-o2(GtA)	-30	0	-23	0	-38	-3	-14	-2	4	-9	3	-1.22	-19	-1	-43	-
ILLUME+(direct)	0	-2	10	2	0	-8	-13	3	2	-14	1	-4.29	-18	-19	2	-
ILLUME+(GtA)	-8	8	9	-2	-4	-13	-17	2	9	-15	3	-6.75	-33	-12	-20	-
gpt4o	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
gpt4o	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GPT-4o + GPT-image	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
gemini pro	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
qwenedit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SenseNova-U1.5-8B-MoT (Direct, unpaired)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SenseNova-U1.5-8B-MoT (Visual CoT, unpaired)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

Table 14: **Delta results on UniG2U across categories (Sheet 3, Part 2/2).** Continuing from Part 1 with the remaining columns. A dash denotes either a missing value or an unavailable delta; SenseNova-U1.5 is unpaired and therefore has no delta.

Model	Geometry Reasoning			Spatial Intl.				Puzzles & Games				Physics			
	Geometry	3d Geometry	MSR	Attr. (Meas.)	Attr. (Appr.)	Motion (Cam.)	Motion (Obj.)	Mental Recon.	Mental Track.	Visual Track.	maze	jigsaw	sliding	light	mechanism
qwen2.5-vl-7b*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
OmniGen2(direct)	-18	0	-3	13	9	-17	0	-5	-2	-2.41	-9.81	-4	-5.61	8	6
OmniGen2(GtA)	-20	0	4	13	7	-11	2	-5	-2	-3.62	-9.52	2	-5.48	7	8
UniWorld-V1(GtA)	-8	0	2	2	6	-8	5	3	-1	-3.62	0.31	-5	0.53	-1	2
OncCAT-3b(direct)	-12	0	7	10	17	-5	-20	-12	0	-3.62	-5.72	-2	5.57	7	-11
OncCAT-3b(GtA)	-29	0	14	8	13	-26	-7	-13	-1	-3.62	-6.69	-2	-2.51	3	-13
JavisGPT(direct)	-4	0	-9	4	8	-10	-2	-6	8	-0.9	-13.64	-1	4.83	-9	-6
uni-video(GtA)	-3	0	-2	-2	7	-19	-1	-8	1	0	7.78	5	25.18	-3	-7
UniPic2(GtA)	-13	1	-4	3	11	-6	2	-3	-3	-4.82	-2.59	-4	13.1	-6	0
STAR-7B(GtA)	-17	0	2	2	7	-8	6	-5	2	-3.62	-7.25	-6	14.11	0	-3
Qwen2.5-VL-3B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
UAE(direct)	-10	0	4	-10	2	2	-4	2	-5	1.21	3.53	-12	5.91	-7	-13
Ovis-U1(direct)	16	0	-10	-4	-17	0	-13	0	4	4.82	11.8	-1	33.39	-7	-13
Ovis-U1(GtA)	2	0	5	-5	-9	-7	-14	-9	-5	0	-1.39	2	0.74	-23	-24
yi-6B-vl*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
miu(direct)	0	0	-3	-2	2	12	-5	-8	-2	-2.41	0	10	0	-5	0
miu(GtA)	0	0	11	0	-10	-1	-9	2	-3	0	0	-20	0	-3	-3
deepseek-v2*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Janus-Pro(direct)	6	0	5	18	16	2	-7	10	25	-2.41	1.33	1	-20.95	0	-6
ALA(direct)	8	0	0	24	12	5	5	20	18	-1.2	1.67	-2	-8.89	10	8
LLaDA-Instruct*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MMaDA(direct)	7	0	11	-14	7	9	-5	-1	2	-1.2	-12.87	-13	0	4	7
Janus-1.3B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
FUDOK(direct)	-3	0	14	0	15	11	-4	-2	6	1.2	-1.81	-18	-0.26	6	-22
Qwen3-VL-8B*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
MammothMod2(direct)	-67	-11	-9	-16	-4	-2	10	3	-2	-1.2	-15.53	-45	-8.59	3	-4
llava-h*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Emu3(direct)	2	0	3	-1	-2	0	-7	7	7	0	1.39	-50	8.5	4	-22
llava_onevision*	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
X-Omni(direct)	-2	0	3	5	-3	4	-1	-1	4	-10.85	5.41	9	1.98	-7	-22
bagel(direct)	37	0	9	5	-9	6	4	6	7	-9.64	-5.31	1	-8.35	-7	-6
bagel(GtA)	37	2	8	10	-2	4	-2	10	8	-12.05	20.64	4	10.2	-4	-3
TokenFlow-XL(direct)	21	0	-1	-2	-11	-1	-4	-1	10	-8.13	-6.98	0	1.19	-17	-18
show-o2(direct)	1	0	10	-2	-8	8	-6	1	-1	-7.23	6.78	5	4.49	-5	-3
show-o2(GtA)	-3	0	13	-11	-29	3	-1	-1	0	-6.03	10.38	1	9.07	1	-9
ILLUME+(direct)	-6	0	11	3	-15	-6	-4	-4	-4	-10.85	-0.53	2	-5.23	-4	-14
ILLUME+(GtA)	-17	0	12	2	-14	-4	-18	-11	-5	-14.46	5.41	1	-9.25	-5	-12
gpt4o	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ovis2.5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GPT-4o + GPT-image	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
gemini pro	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
qwenedit	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SenseNova-U1.5-8B-MoT (Direct, unpaired)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
SenseNova-U1.5-8B-MoT (Visual CoT, unpaired)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

option's letter from the given choices directly. Enclose the option's letter within ``."

G.1.2 MMSI_MOTION_CAM (MOTION - CAMERA) AND MMSI_MOTION_OBJ (MOTION - OBJECT)

Note: These two tasks currently share identical prompts in the non-CoT version.

pre_prompt: "Analyze the consecutive first-person perspective images. Determine how the camera (viewpoint) is moving or rotating by observing changes in the scene and object positions relative to the viewer.\n\n"

post_prompt: "\n\nBased on your analysis of camera motion, answer with the option's letter from the given choices directly. Enclose the option's letter within ``."

G.2 ILLUSIONBENCH TASKS

G.2.1 ICON SHAPE RECOGNITION

shape_string = "Animal, Face_Emoji, Music, Sport, Stationery, Vehicle"
scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune, Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market, Time_square"

prompt = f"This image contains a icon integrated into a background, where elements of the background contribute to forming the icon. Identify the icon that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string} Provide your response by stating only the single, most accurate class name that represents the icon. You have to respond with a single word."

G.2.2 ICON SCENE RECOGNITION, LOGO SCENE RECOGNITION, AND IN SCENE RECOGNITION

Note: These three tasks share identical prompts (only differ in shape_string values).

prompt = f"This image contains an icon integrated into a background, where elements of the background contribute to forming the icon. Identify the background that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string}. Provide your response by stating only the single, most accurate class name that represents the background. You have to respond with a single word."

G.2.3 LOGO SHAPE RECOGNITION

shape_string = "Adidas, Amazon, Apple, Audi, BMW, Mercedes Benz, Facebook, Google, Instagram, Mcdonalds, Nasa, Nike, Olympics, Playstation, Puma, Reebok, Spotify, Starbucks, Tesla, Telegram, Ubuntu"
scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune, Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market, Time_square"

prompt = f"This image contains a icon integrated into a background, where elements of the background contribute to forming the logo. Identify the logo that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string} Provide your response by stating only the single, most accurate class name that represents the logo. You have to respond with a single word."

G.2.4 IN SHAPE RECOGNITION

shape_string = "Airplane, Bicycle, Bird, Bottle, Car, Cat, Dog,

Dolphin, Fork, Guitar, Mug, Panda, Paper_clip, Sailboat, Scooter, Teapot"

scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune, Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market, Time_square"

prompt = f"This image contains a icon integrated into a background, where elements of the background contribute to forming the icon. Identify the shape that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string} Provide your response by stating only the single, most accurate class name that represents the icon. You have to respond with a single word."

G.3 VISUALPUZZLES TASKS

G.3.1 ALL CATEGORIES (ALGORITHMIC, ANALOGICAL, DEDUCTIVE, INDUCTIVE, SPATIAL)

MULTI_CHOICE_DIRECT_PROMPT = "Answer the question with the option's letter from the given choices directly."

Full prompt format:

```
question = "Question: " + doc["question"].strip()
options = doc["options"]
if options is not None:
    question += "\nOptions:\n(A) " + options[0] + "\n(B) " + options[1]
    + "\n(C) " + options[2] + "\n(D) " + options[3]
else:
    question += "\nOptions: Choose from (A) (B) (C) (D) in the image."
question += "\n" + MULTI_CHOICE_DIRECT_PROMPT
```

G.4 UNI-MMMU TASKS

G.4.1 UNI_MMMU_JIGSAW100_VISUAL_COT

Full Prompt:

You are a unified vision-language model. You will be given:

- (1) a 2x2 reference image with the bottom-right cell hidden, and
- (2) two candidate patch images ("Candidate 0" and "Candidate 1").

Your job:

- For each candidate, synthesize a completed 2x2 image by placing that candidate EXACTLY into the bottom-right cell. Keep the other three cells pixel-identical to the reference (no filtering, no re-rendering). If sizes differ, only scale the candidate to fit that quadrant; do NOT rotate, mirror, or alter colors.
- Compare the two completed results and decide which candidate yields the correct completion.

Output EXACTLY the following, in order:

- 1) A single image with Candidate 0 placed in the bottom-right cell
- 2) A single image with Candidate 1 placed in the bottom-right cell
- 3) analysis comparing seam continuity, color/texture gradient, structural alignment, and global semantics
- 4) One strict JSON object with your decision, wrapped as:

```

2106 <FINAL_ANSWER_JSON>
2107 {"choice": 0 or 1, "rationale": "<=30 words decisive cue"}
2108 </FINAL_ANSWER_JSON>
2109
2110 Hard constraints:
2111 - Deterministic, single outcome. No hedging, no multiple possibilities.
2112 - No meta talk about prompts, models, or pipelines.
2113 - Do not restate the task as the answer; reason from visual evidence.
2114 - The only edits allowed are pasting the candidate into the bottom-
2115   right cell and necessary size matching for that cell. All other
2116   pixels must remain identical to the reference.
2117
2118 Inputs:
2119
2120 G.4.2 UNI_MMMU_MAZE100_VISUAL_COT
2121
2122 Full Prompt:
2123
2124 You are a precise maze solver.
2125
2126 SEMANTICS (for all mazes)
2127 - Black squares: walls (impassable)
2128 - White squares: path (walkable)
2129 - Blue dot: start (the agent)
2130 - Green rectangular frame: goal (reaching any white cell inside the
2131   green frame counts as success)
2132 - Legal moves: up, down, left, right only. One cell per step; no
2133   diagonals, no jumps; never cross walls.
2134
2135 OUTPUT FORMAT (STRICT)
2136 1) MULTI-IMAGE MODE - generate a SEQUENCE OF SEPARATE IMAGES, one per
2137   move:
2138   - Each output image must depict the maze state AFTER applying
2139     exactly one legal move.
2140   - Do NOT include the initial (pre-move) state.
2141   - Keep palette/layout/scale identical to the input; only the blue
2142     dot moves.
2143   - The number of returned images MUST equal the number of moves in
2144     the final answer (see step 2).
2145   - Absolutely FORBIDDEN: any collage/montage/spritesheet/grid/
2146     multi-panel/side-by-side/stacked images; no arrows, captions, or
2147     overlays; no GIFs/animations/video.
2148
2149 2) After all step images, emit EXACTLY ONE LINE containing ONLY the
2150   final move list as a JSON array of lowercase strings, wrapped as:
2151   <ANSWER_JSON>["right", "down", "left"]</ANSWER_JSON>
2152
2153 NO EXTRAS
2154 - No tools, no OCR, no explanations, and no text other than the single
2155   <ANSWER_JSON>.....</ANSWER_JSON> line.
2156 - Do not restate the instructions or the condition.
2157
2158 REMINDERS
2159 - Decide the full path first, then emit the image sequence (one image
2160   per move), then the single <ANSWER_JSON> line.
2161 - One move per image; images must be separate files/parts, not stitched
2162   together in any way.
2163
2164 G.4.3 UNI_MMMU_SLIDING54_VISUAL_COT
2165
2166 Full Prompt:
2167
2168 You are a precise sliding puzzle solver.

```

2160 SEMANTICS
 2161 - The puzzle is a 3x3 grid with 8 colored tiles and one empty space
 2162 (shown as red).
 2163 - A "move" slides an adjacent tile INTO the empty space.
 2164 - Moves are named by the direction the COLORED TILE moves (not the
 2165 empty space).
 2166 - Legal moves: up, down, left, right only. One tile per step.

2167 OUTPUT FORMAT (STRICT)
 2168 1) MULTI-IMAGE MODE - generate a SEQUENCE OF SEPARATE IMAGES, one per
 2169 move:
 2170 - Each output image must depict the puzzle state AFTER applying
 2171 exactly one move.
 2172 - Do NOT include the initial (pre-move) state.
 2173 - Keep tile colors identical; only positions change.
 2174 - The number of returned images MUST equal the number of moves in
 2175 the final answer.
 2176 - Absolutely FORBIDDEN: any collage/montage/spritesheet/grid/
 multi-panel/side-by-side/stacked images.

2177 2) After all step images, emit EXACTLY ONE LINE containing ONLY the
 2178 final move list as a JSON array of lowercase strings, wrapped as:
 2179 <ANSWER_JSON>["down","right","up"]</ANSWER_JSON>

2180 NO EXTRAS
 2181 - No explanations beyond the single <ANSWER_JSON>.....</ANSWER_JSON>
 2182 line.
 2183 - Do not restate the instructions.

2185 H TASK PROMPTS (VISUAL COT VERSION)

2187 This appendix contains all the original English prompts for Visual CoT tasks.

2188 All Visual CoT tasks use a two-stage format:

2190 [GEN_PROMPT]...[/GEN_PROMPT][QUESTION]...[/QUESTION]
 2191

2192 H.1 MMSI-BENCH TASKS

2193 H.1.1 MMSI_ATTRIBUTE_APPR_VISUAL_COT

2195 Stage 1 - Generation Prompt:

2196
 2197 Create a visualization that highlights and labels the visual appearance
 2198 attributes (color, shape, texture, orientation, count) in the scene.
 2199 Use annotations, bounding boxes, and labels to make object features and
 2200 counts clearly visible.

2201 Stage 2 - Question Prompt:

2202
 2203 pre_prompt: "You are given the original image(s) and a visualization
 2204 highlighting appearance attributes. Use both to analyze color, shape,
 2205 texture, orientation, and object counts.\n\n"

2206
 2207 post_prompt: "\n\nBased on your visual observation, answer with the
 2208 option's letter from the given choices directly. Enclose the option's
 letter within ``."

2210 H.1.2 MMSI_ATTRIBUTE_MEAS_VISUAL_COT

2211 Stage 1 - Generation Prompt:

2212
 2213 Create a visualization that highlights and annotates the measurable
 attributes (size, length, width, height, distance, area, volume) in

the scene. Draw measurement lines, labels, and comparison markers to make quantitative relationships clear.

Stage 2 - Question Prompt:

pre_prompt: "You are given the original image(s) and a visualization highlighting measurable attributes. Use both to analyze size, length, width, height, distance, area, and volume relationships.\n\n"

post_prompt: "\n\nBased on your measurement analysis, answer with the option's letter from the given choices directly. Enclose the option's letter within ``."

H.1.3 MMSI_MOTION_CAM_VISUAL_COT

Stage 1 - Generation Prompt:

Create a visualization showing camera motion analysis. Draw arrows indicating the direction of camera movement/rotation, highlight reference points that shift between frames, and annotate the type of camera motion (pan, tilt, zoom, dolly, etc.).

Stage 2 - Question Prompt:

pre_prompt: "You are given consecutive first-person perspective images and a visualization of camera motion. Use both to determine how the camera (viewpoint) is moving or rotating.\n\n"

post_prompt: "\n\nBased on your analysis of camera motion, answer with the option's letter from the given choices directly. Enclose the option's letter within ``."

H.1.4 MMSI_MOTION_OBJ_VISUAL_COT

Stage 1 - Generation Prompt:

Create a visualization showing object motion tracking. Draw motion trails, arrows indicating direction of movement, and highlight moving objects with bounding boxes. Annotate relative speeds and trajectories of different objects.

Stage 2 - Question Prompt:

pre_prompt: "You are given consecutive images and a visualization of object motion. Use both to track objects, identify which are moving, their direction, and relative speeds.\n\n"

post_prompt: "\n\nBased on your analysis of object motion, answer with the option's letter from the given choices directly. Enclose the option's letter within ``."

H.1.5 MMSI_MSR_VISUAL_COT

Stage 1 - Generation Prompt:

Create a visualization that breaks down the multi-step reasoning process. Annotate spatial relationships, object positions, and key changes across images. Draw diagrams showing logical connections and intermediate reasoning steps.

Stage 2 - Question Prompt:

pre_prompt: "You are given the original images and a visualization breaking down the reasoning steps. Use both to analyze spatial

relationships, object positions, and changes across images.\n\n"

post_prompt: "\n\nBased on your multi-step reasoning, answer with the option's letter from the given choices directly. Enclose the option's letter within ``."

H.2 ILLUSIONBENCH TASKS

H.2.1 ICON SHAPE RECOGNITION VISUAL CoT

Stage 1 - Generation Prompt:

Create a visualization that highlights how the background elements contribute to forming the icon shape. Draw outlines, arrows, and annotations showing which parts of the scene create the icon's silhouette or features.

Stage 2 - Question Prompt:

shape_string = "Animal, Face_Emoji, Music, Sport, Stationery, Vehicle"
scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune, Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market, Timesquare"

pre_prompt: "You are given the original image and a visualization highlighting how background elements form the icon. Use both to identify the icon shape.\n\n"

prompt = f"This image contains an icon integrated into a background, where elements of the background contribute to forming the icon. Identify the icon that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string} Provide your response by stating only the single, most accurate class name that represents the icon. You have to respond with a single word."

H.2.2 ICON SCENE RECOGNITION VISUAL CoT, LOGO SCENE RECOGNITION VISUAL CoT, AND IN SCENE RECOGNITION VISUAL CoT

Note: These three tasks share identical prompts (only differ in shape_string values).

Stage 1 - Generation Prompt:

Create a visualization that highlights the background scene elements. Draw annotations identifying key features of the environment, landscape, or setting that characterize the scene type.

Stage 2 - Question Prompt:

pre_prompt: "You are given the original image and a visualization highlighting the background scene elements. Use both to identify the scene type.\n\n"

prompt = f"This image contains an icon integrated into a background, where elements of the background contribute to forming the icon. Identify the background that is represented in the image by choosing exclusively among the following options:{shape_string},{scene_string}. Provide your response by stating only the single, most accurate class name that represents the background. You have to respond with a single word."

H.2.3 LOGO SHAPE RECOGNITION VISUAL CoT

Stage 1 - Generation Prompt:

Create a visualization that highlights how the background elements contribute to forming the logo shape. Draw outlines, arrows, and annotations showing which parts of the scene create the logo's silhouette or features.

Stage 2 - Question Prompt:

```
shape_string = "Adidas, Amazon, Apple, Audi, BMW, Mercedes Benz,
Facebook, Google, Instagram, Mcdonalds, Nasa, Nike, Olympics,
Playstation, Puma, Reebok, Spotify, Starbucks, Tesla, Telegram, Ubuntu"
scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune,
Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market,
Time_square"
```

```
pre_prompt: "You are given the original image and a visualization
highlighting how background elements form the logo. Use both to
identify the logo.\n\n"
```

```
prompt = f"This image contains an icon integrated into a background,
where elements of the background contribute to forming the logo.
Identify the logo that is represented in the image by choosing
exclusively among the following options:{shape_string},{scene_string}
Provide your response by stating only the single, most accurate class
name that represents the logo. You have to respond with a single word."
```

H.2.4 IN SHAPE RECOGNITION VISUAL COT

Stage 1 - Generation Prompt:

Create a visualization that highlights how the background elements contribute to forming the shape. Draw outlines, arrows, and annotations showing which parts of the scene create the shape's silhouette or features.

Stage 2 - Question Prompt:

```
shape_string = "Airplane, Bicycle, Bird, Bottle, Car, Cat, Dog,
Dolphin, Fork, Guitar, Mug, Panda, Paper_clip, Sailboat, Scooter,
Teapot"
scene_string = "Ocean, Origami, Forest, Cloud, Sand_dune,
Medieval_Village, City, Underwater_ruins, Museum, Bazaar_market,
Time_square"
```

```
pre_prompt: "You are given the original image and a visualization
highlighting how background elements form the shape. Use both to
identify the shape.\n\n"
```

```
prompt = f"This image contains an icon integrated into a background,
where elements of the background contribute to forming the icon.
Identify the shape that is represented in the image by choosing
exclusively among the following options:{shape_string},{scene_string}
Provide your response by stating only the single, most accurate class
name that represents the icon. You have to respond with a single word."
```

H.3 VISUALPUZZLES TASKS

H.3.1 ALGORITHMIC VISUAL COT

Stage 1 - Generation Prompt:

Create a visualization that breaks down the algorithmic pattern or sequence. Draw step-by-step diagrams, number sequences, or flowcharts showing the logical progression and rules governing the pattern.

Stage 2 - Question Prompt:

```

2376 pre_prompt: "You are given the original puzzle image(s) and a
2377 visualization breaking down the algorithmic pattern. Use both to
2378 identify the underlying algorithm or sequence rule.\n\n"
2379
2380 MULTI_CHOICE_DIRECT_PROMPT = "Answer the question with the option's
2381 letter from the given choices directly."
2382
2383 question = "Question: " + doc["question"].strip()
2384 options = doc["options"]
2385 if options is not None:
2386     question += "\nOptions:\n(A) " + options[0] + "\n(B) " + options[1]
2387     + "\n(C) " + options[2] + "\n(D) " + options[3]
2388 else:
2389     question += "\nOptions: Choose from (A) (B) (C) (D) in the image."
2390
2391 question += "\n" + MULTI_CHOICE_DIRECT_PROMPT

```

H.3.2 ANALOGICAL VISUAL COT

Stage 1 - Generation Prompt:

```

2394 You are given an analogical reasoning puzzle (A is to B as C is to ?).
2395
2396 {question}
2397
2398 Your task:
2399 1. Identify the transformation or relationship from A to B
2400 2. Create a diagram that clearly shows:
2401    - The key features that change from A to B
2402    - The transformation rule (rotation, color change, shape
2403      modification, etc.)
2404    - How the same rule applies to C
2405    - Annotations or arrows showing the transformation steps
2406 3. Make the transformation pattern visually obvious
2407
2408 Generate a diagram that reveals the transformation pattern between
2409 pairs.

```

Stage 2 - Question Prompt:

```

2410 {question}
2411
2412 You are given TWO images:
2413 1) ORIGINAL PUZZLE: The analogical reasoning puzzle
2414 2) AUXILIARY DIAGRAM: A visualization showing the transformation
2415    relationship
2416
2417 Use the auxiliary diagram to understand how A transforms to B, apply
2418 the same rule to C, then select the correct answer.
2419 Answer with the option letter (A, B, C, or D) directly.

```

H.3.3 DEDUCTIVE VISUAL COT

Stage 1 - Generation Prompt:

```

2423 You are given a deductive reasoning puzzle that requires logical
2424 inference.
2425
2426 {question}
2427
2428 Your task:
2429 1. Identify the given premises, rules, or constraints in the puzzle
2430 2. Create a diagram that clearly shows:
2431    - All given conditions/rules listed clearly

```

2430 - A logical flowchart or inference chain
2431 - Step-by-step deduction from premises to conclusion
2432 - Elimination of incorrect possibilities
2433 - The logical path leading to the answer
2434 3. Use arrows to show the deduction flow
2435
2436 Generate a logical inference diagram that traces the reasoning path.
2437
2438 **Stage 2 - Question Prompt:**
2439 {question}
2440
2441 You are given TWO images:
2442 1) ORIGINAL PUZZLE: The deductive reasoning puzzle
2443 2) AUXILIARY DIAGRAM: A logical inference diagram showing the deduction
2444 steps
2445
2446 Follow the logical chain in the auxiliary diagram to reach the
2447 conclusion, then select the correct answer.
2448 Answer with the option letter (A, B, C, or D) directly.
2449
2450 **H.3.4 INDUCTIVE VISUAL CoT**
2451
2452 **Stage 1 - Generation Prompt:**
2453 You are given an inductive reasoning puzzle that requires pattern
2454 recognition.
2455 {question}
2456
2457 Your task:
2458 1. Observe the sequence of examples and identify the underlying pattern
2459 2. Create a diagram that clearly shows:
2460 - The repeating elements or motifs highlighted/circled
2461 - The progression rule (what changes from one step to the next)
2462 - Annotations showing the pattern cycle or growth rule
2463 - A prediction of what comes next based on the pattern
2464 - Color-coding or numbering to show pattern repetition
2465 3. Make the inductive pattern visually obvious
2466
2467 Generate a diagram that highlights the repeating pattern and predicts
2468 the next element.
2469
2470 **Stage 2 - Question Prompt:**
2471 {question}
2472
2473 You are given TWO images:
2474 1) ORIGINAL PUZZLE: The inductive reasoning puzzle
2475 2) AUXILIARY DIAGRAM: A visualization highlighting the pattern and its
2476 progression
2477
2478 Use the auxiliary diagram to identify the pattern, then select the
2479 correct answer.
2480 Answer with the option letter (A, B, C, or D) directly.
2481
2482 **H.3.5 SPATIAL VISUAL CoT**
2483 **Stage 1 - Generation Prompt:**
2484 You are given a spatial reasoning puzzle involving 3D visualization or
2485 transformations.
2486 {question}

2484
2485 Your task:
2486 1. Analyze the spatial transformation required (rotation, folding,
2487 unfolding, different viewpoint)
2488 2. Create a diagram that clearly shows:
2489 - The object from multiple angles if rotation is involved
2490 - Step-by-step folding/unfolding process if applicable
2491 - Arrows indicating rotation direction and degree
2492 - Reference points or markers to track orientation
2493 - The resulting shape after transformation
2494 3. Add axis lines or reference frames to clarify spatial orientation
2495
2496 Generate a multi-view or step-by-step transformation diagram.

2496 **Stage 2 - Question Prompt:**

2497 {question}

2498 You are given TWO images:

- 2499 1) ORIGINAL PUZZLE: The spatial reasoning puzzle
2500 2) AUXILIARY DIAGRAM: A visualization showing the spatial
2501 transformation from multiple views
2502

2503 Use the auxiliary diagram to mentally trace the transformation, then
2504 select the correct answer.

2505 Answer with the option letter (A, B, C, or D) directly.
2506

2507 **H.4 CHARTQA TASKS**

2508 **H.4.1 CHARTQA100_VISUAL_COT**

2509 **Stage 1 - Generation Prompt:**

2510 Based on this chart and the question being asked, generate an annotated
2511 version of the chart that highlights the relevant data points, bars,
2512 lines, or regions needed to answer the question. Mark or circle the key
2513 values and label them clearly.
2514

2515 **Stage 2 - Question Prompt:**

2516 In addition to the original chart, you are also given an annotated
2517 version that highlights the relevant data points for the question.
2518

2519 {question}

2520 Answer the question with a single word.
2521

2522 **H.5 GEOMETRY TASKS**

2523 **H.5.1 GEOMETRY3K_VISUAL_COT**

2524 **Stage 1 - Generation Prompt:**

2525 Based on this geometry problem, generate an auxiliary diagram that adds
2526 helpful constructions to solve the problem.
2527

2528 Your task:

- 2529 1. Analyze what geometric relationships need to be proven or calculated
2530 2. Add auxiliary lines, points, or constructions that reveal these
2531 relationships
2532 3. Label any new points, angles, or segments clearly
2533 4. Highlight key geometric properties (parallel lines, congruent
2534 segments, equal angles, etc.)
2535
2536
2537

Generate a diagram with auxiliary constructions that make the solution path clearer.

Stage 2 - Question Prompt:

You are given TWO images:

- 1) ORIGINAL DIAGRAM: The geometry problem diagram
- 2) AUXILIARY DIAGRAM: The same diagram with helpful auxiliary constructions added

Use both diagrams to solve the problem. The auxiliary constructions should help you identify key relationships.

{question}

H.5.2 AUXSOLIDMATH_EASY_VISUAL_COT

Stage 1 - Generation Prompt:

You are given a solid geometry problem with a 3D diagram. Analyze the problem and create an enhanced version of the SAME diagram with auxiliary constructions added.

Problem: {question}

Instructions:

1. KEEP all original elements exactly as they are (all points, edges, faces, labels)
2. Analyze what auxiliary constructions would help solve this problem
3. ADD auxiliary lines in a different color (e.g., red or dashed lines). Common auxiliary constructions include:
 - Perpendiculars from a point to a plane or line
 - Line segments connecting specific points
 - Midpoints of edges with connections
 - Extended lines to find intersections
 - Parallel lines through specific points
 - Cross-sections of the solid
4. Label any new points you add (use letters not already in the diagram)
5. The final diagram should look like the original 3D figure with extra auxiliary lines drawn on top

Generate an enhanced 3D diagram that preserves the original and adds helpful auxiliary constructions.

Stage 2 - Question Prompt:

You are given a solid geometry problem.

Problem: {question}

You are given TWO images:

- 1) ORIGINAL DIAGRAM: The 3D solid geometry figure as given
- 2) AUXILIARY DIAGRAM: The same figure with auxiliary constructions (extra lines) added to help solve the problem

Instructions:

1. Look at the auxiliary diagram to see what constructions were added (perpendiculars, connecting segments, midpoints, etc.)
2. Identify the geometric relationships established by these auxiliary lines
3. Use these constructions to set up your solution approach
4. Apply relevant theorems (Pythagorean theorem in 3D, properties of perpendiculars, volume formulas, etc.)

5. Show your step-by-step solution with clear calculations
6. State the final numerical answer

Solve this problem step by step using the auxiliary constructions.

H.6 PHYSICS TASKS

H.6.1 PHYX_OPTICS100_VISUAL_COT

Stage 1 - Generation Prompt:

Based on this optics problem, draw a light ray diagram that helps solve the problem. Show the path of light rays through the optical system, including incident rays, reflected rays, refracted rays, focal points, and any relevant angles or focal points as needed by the problem.

Stage 2 - Question Prompt:

In addition to the original image, you are also given an auxiliary light ray diagram to help you solve the problem.

{question}

H.6.2 PHYX_MECHANICS100_VISUAL_COT

Stage 1 - Generation Prompt:

Based on this mechanics problem, draw a free body diagram (force analysis diagram) that helps solve the problem. Show all the forces acting on the object(s), including gravity, normal force, friction, tension, applied forces, etc., with arrows indicating direction and relative magnitude.

Stage 2 - Question Prompt:

In addition to the original image, you are also given an auxiliary free body diagram (force analysis diagram) to help you solve the problem.

{question}

H.7 REALUNIFY GEU TASKS

H.7.1 REALUNIFY_MENTAL_TRACKING_VISUAL_COT

Stage 1 - Generation Prompt:

Here is the question: {question}

Please apply the corresponding transformations and modifications to the contents of the image according to the question.

Stage 2 - Question Prompt:

In addition to the original image, you are also given a transformed version of the image with the modifications applied according to the question.

{prompt}

H.7.2 REALUNIFY_MENTAL_RECONSTRUCTION_VISUAL_COT

Stage 1 - Generation Prompt:

Please restore the image that has been shuffled by patches, without adding extra content or altering the original image.

Stage 2 - Question Prompt:

In addition to the original image, you are also given a restored version of the shuffled image to help you answer the question.

{prompt}

H.7.3 REALUNIFY_ATTENTIONAL_FOCUSING_VISUAL_COT**Stage 1 - Generation Prompt:**

Here is the question: {question}
Please highlight the regions of the image that are relevant to the question.

Stage 2 - Question Prompt:

In addition to the original image, you are also given a visualization that highlights the regions relevant to the question.

{prompt}

H.8 VSP (VISUAL-SPATIAL-PLANNING) TASKS**H.8.1 VSP_GOOGLE_MAP_VISUAL_COT****Stage 1 - Generation Prompt:**

This image shows a map with a starting point marked 'S' (blue) and a goal marked 'G' (red). Your task: Generate a clear visualization that highlights the optimal path from S to G. Draw a simplified map showing the route with arrows indicating directions (North, South, East, West) and mark the number of crossroads to pass in each direction.

Stage 2 - Question Prompt:

In addition to the original images, you are also given an auxiliary visualization image showing the path analysis.

As a professional pathfinder, your task is to analyze a map and find a route from the starting location to the goal. Since coding is not within your skill set, your approach relies on logical reasoning of the map.

Game Setup

- The game presents a fully observable map.
- The starting location is marked with blue "S", and the goal is marked with red "G".
- Your goal is to find a path from the starting location to the goal.

Moving Rules

- The action plan involves moves in four directions: 'W' (West), 'E' (east), 'N' (north), or 'S' (south).
- Each move is along with distances. Distances are measured by how many crossroads passed.

We provide an example to further illustrate the rules.

[Example Image]

In this provided example:

- You are now at the southwest of the goal.
- If you move north by 1 crossroad, you will be at the west of the goal.
- If you move east by 4 crossroads, you will be at the goal.

2700 - IMPORTANT: Please ignore the name of the street and avenue. The
 2701 numbers in the name cannot be used to compute how many crossroads
 2702 need to be passed.

2703 ## Procedure and Output
 2704 Now you will solve the given maze. To analyze the relative spatial
 2705 relation between the starting point and the goal (for example,
 2706 southwest). Then, output a path using the format <Direction>: <Number
 2707 of crossroads passed>.
 2708 For example:
 2709 <Output>
 2710 1. North: 1
 2711 2. East: 4
 2712 means move north by 1 crossroad, and move east by 4 crossroads.
 2713 <Output>
 2714 1. South: 1
 2715 means move south by 1 crossroad.
 2716 Do not output any extra content after the above aggregated output.

2717 Please output path for the following map:
 2718 [Test Image]

2719

2720 H.8.2 VSP_COLLISION_VISUAL_COT

2721 **Stage 1 - Generation Prompt:**

2722

2723 This image shows a map with a car, a person, and a goal. Your task:
 2724 Generate a clear visualization that analyzes the distances and paths.
 2725 Draw a simplified diagram showing the grid distances from the car to
 2726 the goal and from the person to the goal, with arrows indicating their
 2727 movement directions.

2728 **Stage 2 - Question Prompt:**

2729

2730 In addition to the original images, you are also given an auxiliary
 2731 visualization image showing the distance analysis.

2732 As a professional navigation agent, your task is to analyze a map and
 2733 determine the time needed for the car and the person passing the goal.

2734

2735 ## Game Setup

2736 - The game presents a fully observable map. There is a person, a car,
 2737 and a goal on the map.
 2738 - The game further specifies the moving direction of the person and car
 2739 ("up", "down", "left", "right").
 2740 - Your goal is to determine the time needed for the car and the person
 2741 passing the goal.
 2742 The following figure shows how the player, the car, and the goals look
 2743 like.

2744 [Icon Image]

2745 We provide an example to further illustrate the rules.

2746 [Example Image]

2747

2748 The car is moving left with speed 1.0 grid per second, and the person
 2749 is moving up with speed 0.5 grid per second.

2750 In this provided example:

2751 - The car is 2 grid away from the goal. Given it's time as 1.0 grid per
 2752 second, the time needed is $2 / 1.0 = 2$ seconds.
 2753 - The person is 1 grid away from the goal. Given it's time as 0.5 grid
 per second, the time needed is $1 / 0.5 = 2$ seconds.

2754
 2755 ## Procedure and Output
 2756 Now you will answer for the following given map. To solve it, analyze
 2757 the car and the person separately. Then, answer for them separately.
 2758 For example:
 2759 Car: 2.0
 2760 Person: 2.0
 2761 means car and the person will need 2.0 seconds to pass the goal
 2762 respectively.
 2763 Do not output any extra content after the above aggregated output.
 2764
 2765 Please analyze and determine the time needed for the car and the person
 2766 passing the goal:
 2767
 2768 [Test Image]
 2769
 2770 The car is moving {car_dir} with speed {car_speed}, and the person is
 2771 moving {person_dir} with speed {person_speed}.

2772 H.9 BABYVISION TASKS

2773 H.9.1 BABYVISION_FINE_GRAINED_VISUAL_COT

2774 **Stage 1 - Generation Prompt:**
 2775
 2776 This image contains subtle visual details that need careful
 2777 examination. Your task: Generate a visualization that highlights and
 2778 emphasizes the fine-grained details, subtle differences, and
 2779 distinguishing features in the image. Make the key discriminative
 2780 elements more prominent and easier to identify.

2781 **Stage 2 - Question Prompt:**
 2782
 2783 In addition to the original image, you are also given an auxiliary
 2784 visualization image that highlights key visual elements.
 2785 {question}
 2786
 2787 Think step by step and give your final answer in \boxed{Answer} format.

2788 H.9.2 BABYVISION_VISUAL_TRACKING_VISUAL_COT

2789 **Stage 1 - Generation Prompt:**
 2790
 2791 This image involves tracking objects or their movements. Your task:
 2792 Generate a visualization that highlights the objects of interest, their
 2793 positions, trajectories, or movement patterns. Make it easier to follow
 2794 and track the relevant visual elements.

2795 **Stage 2 - Question Prompt:**
 2796
 2797 In addition to the original image, you are also given an auxiliary
 2798 visualization image that highlights key visual elements.
 2799 {question}
 2800
 2801 Think step by step and give your final answer in \boxed{Answer} format.

2802 H.10 UNI-MMMU TASKS

2803 H.10.1 UNI_MMMU_JIGSAW100_VISUAL_COT

2804 **Full Prompt:**

2808 You are a unified vision-language model. You will be given:
 2809 (1) a 2x2 reference image with the bottom-right cell hidden, and
 2810 (2) two candidate patch images ("Candidate 0" and "Candidate 1").
 2811
 2812 Your job:
 2813 - For each candidate, synthesize a completed 2x2 image by placing that
 2814 candidate EXACTLY into the bottom-right cell. Keep the other three
 2815 cells pixel-identical to the reference (no filtering, no
 2816 re-rendering). If sizes differ, only scale the candidate to fit that
 2817 quadrant; do NOT rotate, mirror, or alter colors.
 2818 - Compare the two completed results and decide which candidate yields
 2819 the correct completion.
 2820
 2821 Output EXACTLY the following, in order:
 2822 1) A single image with Candidate 0 placed in the bottom-right cell
 2823 2) A single image with Candidate 1 placed in the bottom-right cell
 2824
 2825 3) analysis comparing seam continuity, color/texture gradient,
 2826 structural alignment, and global semantics
 2827
 2828 4) One strict JSON object with your decision, wrapped as:
 2829 <FINAL_ANSWER_JSON>
 2830 {"choice": 0 or 1, "rationale": "<=30 words decisive cue"}
 2831 </FINAL_ANSWER_JSON>
 2832
 2833 Hard constraints:
 2834 - Deterministic, single outcome. No hedging, no multiple possibilities.
 2835 - No meta talk about prompts, models, or pipelines.
 2836 - Do not restate the task as the answer; reason from visual evidence.
 2837 - The only edits allowed are pasting the candidate into the bottom-
 2838 right cell and necessary size matching for that cell. All other
 2839 pixels must remain identical to the reference.
 2840
 2841 Inputs:
 2842
 2843 H.10.2 UNI_MMMU_MAZE100_VISUAL_COT
 2844
 2845 **Full Prompt:**
 2846
 2847 You are a precise maze solver.
 2848
 2849 SEMANTICS (for all mazes)
 2850 - Black squares: walls (impassable)
 2851 - White squares: path (walkable)
 2852 - Blue dot: start (the agent)
 2853 - Green rectangular frame: goal (reaching any white cell inside the
 2854 green frame counts as success)
 2855 - Legal moves: up, down, left, right only. One cell per step; no
 2856 diagonals, no jumps; never cross walls.
 2857
 2858 OUTPUT FORMAT (STRICT)
 2859 1) MULTI-IMAGE MODE - generate a SEQUENCE OF SEPARATE IMAGES, one per
 2860 move:
 2861 - Each output image must depict the maze state AFTER applying
 2862 exactly one legal move.
 2863 - Do NOT include the initial (pre-move) state.
 2864 - Keep palette/layout/scale identical to the input; only the blue
 2865 dot moves.
 2866 - The number of returned images MUST equal the number of moves in
 2867 the final answer (see step 2).
 2868 - Absolutely FORBIDDEN: any collage/montage/spritesheet/grid/
 2869 multi-panel/side-by-side/stacked images; no arrows, captions, or

overlays; no GIFs/animations/video.

2) After all step images, emit EXACTLY ONE LINE containing ONLY the final move list as a JSON array of lowercase strings, wrapped as: `<ANSWER_JSON>["right","down","left"]</ANSWER_JSON>`

NO EXTRAS

- No tools, no OCR, no explanations, and no text other than the single `<ANSWER_JSON>.....</ANSWER_JSON>` line.
- Do not restate the instructions or the condition.

REMINDERS

- Decide the full path first, then emit the image sequence (one image per move), then the single `<ANSWER_JSON>` line.
- One move per image; images must be separate files/parts, not stitched together in any way.

H.10.3 UNI_MMMU_SLIDING54_VISUAL_COT

Full Prompt:

You are a precise sliding puzzle solver.

SEMANTICS

- The puzzle is a 3x3 grid with 8 colored tiles and one empty space (shown as red).
- A "move" slides an adjacent tile INTO the empty space.
- Moves are named by the direction the COLORED TILE moves (not the empty space).
- Legal moves: up, down, left, right only. One tile per step.

OUTPUT FORMAT (STRICT)

1) MULTI-IMAGE MODE - generate a SEQUENCE OF SEPARATE IMAGES, one per move:

- Each output image must depict the puzzle state AFTER applying exactly one move.
- Do NOT include the initial (pre-move) state.
- Keep tile colors identical; only positions change.
- The number of returned images MUST equal the number of moves in the final answer.
- Absolutely FORBIDDEN: any collage/montage/spritesheet/grid/multi-panel/side-by-side/stacked images.

2) After all step images, emit EXACTLY ONE LINE containing ONLY the final move list as a JSON array of lowercase strings, wrapped as: `<ANSWER_JSON>["down","right","up"]</ANSWER_JSON>`

NO EXTRAS

- No explanations beyond the single `<ANSWER_JSON>.....</ANSWER_JSON>` line.
- Do not restate the instructions.

I RA AND AL EVALUATION PIPELINE AND FULL PROMPT SPECIFICATION

I.1 PURPOSE

This appendix documents the complete evaluation protocol used for Visual Chain-of-Thought (Visual CoT) quality assessment with two judge metrics:

- **RA** (Reasoning-to-Visual Alignment)
- **AL** (Answer-to-Visual Alignment)

The implementation is fully prompt-based and uses a multimodal LLM judge.

I.2 PER-SAMPLE INPUTS AND METADATA

For each sample i , the evaluator reads:

- Original image(s): $I_{\text{orig}}^{(i)}$
- Generated auxiliary image(s): $I_{\text{aux}}^{(i)}$
- Generation prompt: $p_{\text{gen}}^{(i)}$
- Original question: $q^{(i)}$
- Final Stage-2 answer: $a^{(i)}$
- Task name: $t^{(i)}$

The metadata file contains:

```
{
  "doc_id": ...,
  "task": ...,
  "generation_prompt": ...,
  "generated_images": [...],
  "question": ...,
  "stage2_answer": ...
}
```

I.3 METRIC DEFINITIONS

RA.

$$\text{RA}_i = \mathcal{J}_{\text{RA}}(I_{\text{orig}}^{(i)}, I_{\text{aux}}^{(i)}, p_{\text{gen}}^{(i)}, q^{(i)}, t^{(i)}) \in \{1, 2, 3, 4, 5\}$$

RA evaluates whether generated auxiliary visualization follows the generation instruction and is useful/relevant.

AL.

$$\text{AL}_i = \mathcal{J}_{\text{AL}}(I_{\text{orig}}^{(i)}, I_{\text{aux}}^{(i)}, q^{(i)}, a^{(i)}, t^{(i)}) \in \{1, 2, 3, 4, 5\}$$

AL evaluates whether the final answer is consistent with visual evidence and question intent.

Dataset-level averages. For N samples:

$$\overline{\text{RA}} = \frac{1}{N} \sum_{i=1}^N \text{RA}_i, \quad \overline{\text{AL}} = \frac{1}{N} \sum_{i=1}^N \text{AL}_i.$$

I.4 END-TO-END EVALUATION FLOW

For each sample:

1. Load metadata JSON.
2. Resolve `generated_images` (relative paths are optionally resolved via `base_dir`).
3. Construct RA prompt: Base RA Prompt + Category-specific RA Addition.
4. Call multimodal judge with images $[I_{\text{orig}}, I_{\text{aux}}]$, parse `reasoning_visual_score`.
5. Construct AL prompt: Base AL Prompt + Category-specific AL Addition.
6. Call multimodal judge with images $[I_{\text{orig}}, I_{\text{aux}}]$, parse `answer_visual_score`.
7. Save output fields: `ra_score`, `ra_reasoning`, `al_score`, `al_reasoning`.

I.5 JUDGE CALL SETTINGS

- Multimodal judge: GPT-4o compatible endpoint (Azure OpenAI interface).
- Temperature: 0.0.
- Max response tokens: 3000.
- Retry strategy: metric-level retries and API-level retries with backoff.
- Image encoding: JPEG base64 payload.

I.6 PROMPT COMPOSITION RULE

Prompt construction uses this priority:

1. If `task_category` is explicitly set, append that category's criteria.
2. Else, apply legacy task-name mapping:
 - `illusionbench` → `perception`
 - `chartqa` → `chart_table`
 - `mathvista` → `mathematical`
3. Else, use base prompt only.

I.7 COMPLETE BASE PROMPT: RA

You are a professional AI evaluation specialist for Visual Chain-of-Thought tasks.

Task Context

- `**Task**`: {task}
- `**Original Question**`: {question}

Your Objective

Evaluate if the `**generated visualization images**` correctly follow the `**generation prompt**`.

Images Provided

- `**Image 1**`: Original question image (the input)
- `**Image 2**`: Generated visualization image(s) (the auxiliary output created to help answer the question)

Generation Prompt (What model was instructed to generate)

"{generation_prompt}"

Evaluation Criteria

1. Instruction Following (40%)

- Does the generated image follow the generation prompt instructions?
- Are all requested visual elements present?
- Does it match the specified visualization type?

2. Visual Quality (30%)

- Is the generated image clear and well-formed?
- Are visual elements properly rendered?
- Is it helpful as an auxiliary visualization?

3. Relevance to Task (30%)

- Does the visualization help answer the original question?
- Is it relevant to the task context?
- Does it provide useful visual cues?

Scoring Scale (1-5)

- `**5 Perfect**`: Generated image(s) perfectly match generation prompt; all instructions followed; high quality; highly relevant
- `**4 Good**`: Generated image(s) mostly match prompt (80-90%); minor issues; good quality; relevant
- `**3 Adequate**`: Generated image(s) partially match prompt (60-70%); noticeable issues; acceptable quality
- `**2 Poor**`: Generated image(s) barely match prompt (30-50%); major issues; poor quality or relevance
- `**1 Failed**`: Generated image(s) completely fail to match prompt; unusable; irrelevant

Important Notes

- Focus on whether the `**generation prompt**` was correctly executed

```

3024 - Do NOT evaluate if the final answer is correct (that's a separate metric)
3025 - Evaluate the visualization quality and instruction adherence
3026
3027 ## Output Format
3028 {
3029   "reasoning_visual_score": X,
3030   "reasoning": "Detailed evaluation of how well generated images match the
3031   generation prompt"
3032 }
3033
3034 I.8 COMPLETE BASE PROMPT: AL
3035
3036 You are a professional AI evaluation specialist for Visual Chain-of-Thought tasks.
3037
3038 ## Task Context
3039 - Task: {task}
3040 - Original Question: {question}
3041
3042 ## Your Objective
3043 Evaluate if the final answer is consistent with the generated visualization
3044 images and the original question.
3045
3046 ## Images Provided
3047 - Image 1: Original question image (the input)
3048 - Image 2: Generated visualization image(s) (auxiliary output used to
3049   derive the answer)
3050
3051 ## Final Answer
3052 "{answer}"
3053
3054 ## Evaluation Criteria
3055
3056 ### 1. Visual-Answer Consistency (50%)
3057 - Is the answer supported by visual evidence in generated images?
3058 - Can you trace the answer back to visual elements?
3059 - Are there contradictions between answer and visuals?
3060
3061 ### 2. Question-Answer Alignment (30%)
3062 - Does the answer correctly address the original question?
3063 - Is the answer format appropriate?
3064 - Is it a valid response to the question?
3065
3066 ### 3. Reasoning Coherence (20%)
3067 - Does the reasoning flow make sense (original → generated visual → answer)?
3068 - Is the answer logically derived from the visualization?
3069 - Is the overall chain-of-thought coherent?
3070
3071 ## Scoring Scale (1-5)
3072
3073 - 5 Perfect: Answer perfectly matches generated visuals and question;
3074   fully consistent; logically sound; correct reasoning
3075 - 4 Good: Answer mostly consistent (80-90%);
3076   minor inconsistencies; generally correct reasoning
3077 - 3 Adequate: Answer partially consistent (60-70%);
3078   some contradictions; reasoning has gaps
3079 - 2 Poor: Answer barely consistent (30-50%);
3080   major contradictions; flawed reasoning
3081 - 1 Failed: Answer completely inconsistent with visuals or question;
3082   contradictory; nonsensical
3083
3084 ## Important Notes
3085 - Focus on consistency between answer and generated visuals
3086 - Check if answer logically follows from the visualization
3087 - Verify answer addresses the original question

```

```

- Do NOT evaluate if generation prompt was followed (that's RA metric)

## Output Format
{
  "answer_visual_score": X,
  "reasoning": "Detailed evaluation of answer consistency with visuals and question"
}

```

I.9 FULL CATEGORY-SPECIFIC ADDITIONS

A. REAL-WORLD APPLICATIONS (REAL_WORLD)

RA addition

```

## Real-world Applications - RA Criteria
- **Safety & Feasibility**: Does visualization show safe, practical steps?
- **Contextual Understanding**: Does it consider real-world constraints
  (time, resources, safety)?
- **Actionable Guidance**: Are visual steps clear enough to follow in practice?
- **Ethical Considerations**: Does visualization respect safety protocols and
  ethical guidelines?

```

AL addition

```

## Real-world Applications - AL Criteria
- **Practical Validity**: Is the answer practically feasible and safe?
- **Evidence-Based**: Is answer supported by visual evidence showing real-world execution?
- **Completeness**: Does answer address all critical real-world aspects
  (safety, resources)?
- **Ethical Soundness**: Is the answer ethically appropriate and safe?

```

B. GEOMETRY REASONING (MATHEMATICAL)

RA addition

```

## Mathematical Reasoning - RA Criteria
- **Formula Visualization**: Are mathematical formulas and equations
  properly visualized?
- **Step-by-Step Clarity**: Does visualization show clear calculation steps?
- **Numerical Accuracy**: Are all numerical values and calculations visually correct?
- **Mathematical Notation**: Is proper mathematical notation used?

```

AL addition

```

## Mathematical Reasoning - AL Criteria
- **Calculation Correctness**: Is the answer mathematically accurate?
- **Formula Application**: Does answer correctly apply formulas shown in visualization?
- **Logical Consistency**: Is the mathematical logic coherent from visual → answer?
- **Unit Consistency**: Are units and scales properly maintained?

```

C. PHYSICS (STEM)

RA addition

```

## Physics - RA Criteria
- **Scientific Accuracy**: Does visualization follow scientific principles and laws?
- **Technical Precision**: Are technical details and processes accurately depicted?
- **Domain Knowledge**: Does it demonstrate correct Physics domain knowledge?
- **Experimental/Process Visualization**: Are processes or experiments
  properly illustrated?

```

AL addition

```

## Physics - AL Criteria

```

- **Scientific Validity**: Is answer scientifically correct?
- **Technical Accuracy**: Does answer align with technical/scientific principles shown?
- **Evidence-Based**: Is answer supported by scientific evidence in visualization?
- **Domain Consistency**: Does answer demonstrate proper Physics domain understanding?

D. PUZZLES AND GAMES (PUZZLES)

RA addition

Puzzles and Games - RA Criteria

- **Rule Adherence**: Does visualization follow game/puzzle rules correctly?
- **Strategic Clarity**: Are strategic steps or moves clearly visualized?
- **Valid Moves**: Are all shown moves/steps valid according to rules?
- **Solution Path**: Does visualization show a valid path to solution?

AL addition

Puzzles and Games - AL Criteria

- **Rule Compliance**: Does answer follow all game/puzzle rules?
- **Solution Validity**: Is the answer a valid solution to the puzzle?
- **Move Correctness**: Do final moves/steps match what's shown in visualization?
- **Win Condition**: Does answer achieve the stated goal/win condition?

E. CHART & TABLE REASONING (CHART_TABLE)

RA addition

Chart & Table Reasoning - RA Criteria

- **Data Highlighting**: Are relevant data points properly highlighted?
- **Trend Visualization**: Are trends, patterns, or key values emphasized?
- **Label Clarity**: Are axes, labels, and legends clearly marked?
- **Data Accuracy**: Is extracted/highlighted data numerically accurate?

AL addition

Chart & Table Reasoning - AL Criteria

- **Data Accuracy**: Does answer match actual data values in chart/table?
- **Trend Correctness**: If describing trends, is interpretation correct?
- **Numerical Precision**: Are numerical answers precise and accurate?
- **No Hallucination**: Is answer derived from actual data, not hallucinated?

F. SPATIAL INTELLIGENCE (SPATIAL)

RA addition

Spatial Intelligence - RA Criteria

- **3D Understanding**: Does visualization correctly represent 3D spatial relationships?
- **Transformation Accuracy**: Are rotations, translations, or transformations correct?
- **Perspective Consistency**: Is spatial perspective maintained correctly?
- **Geometric Precision**: Are geometric relationships accurately visualized?

AL addition

Spatial Intelligence - AL Criteria

- **Spatial Correctness**: Is answer spatially accurate?
- **Transformation Validity**: Do described transformations match visualization?
- **Geometric Consistency**: Is answer geometrically consistent with shown spatial relationships?
- **Orientation Accuracy**: Are orientations and positions correctly described?

G. PERCEPTION REASONING (PERCEPTION)

RA addition

```

## Perception Reasoning - RA Criteria
- **Object Highlighting**: Are relevant objects properly highlighted or marked?
- **Scene Understanding**: Does visualization show correct scene interpretation?
- **Feature Emphasis**: Are key perceptual features emphasized?
- **Visual Clarity**: Is the visualization clear for perception tasks?

```

AL addition

```

## Perception Reasoning - AL Criteria
- **Object Identification**: Are identified objects correct?
- **Scene Accuracy**: Does answer accurately describe the scene?
- **Attribute Correctness**: Are object attributes (color, shape, position) correct?
- **Visual Evidence**: Is answer supported by visible evidence in images?

```

I.10 SCORING OUTPUTS AND SCHEMA

Per-sample output:

```

{
  "doc_id": "...",
  "task": "...",
  "json_path": "...",
  "ra_score": 1.5 or null,
  "ra_reasoning": "...",
  "al_score": 1.5 or null,
  "al_reasoning": "..."
}

```

I.11 NOTES ON ROBUSTNESS

- If JSON parsing of judge response fails, regex extraction is used as fallback.
- If all retries fail, score is returned as null.
- Multi-image tasks are supported by concatenating all original images before auxiliary images.